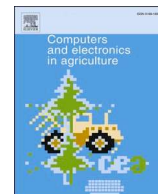




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Original papers

LGENetB4CA: A novel deep learning approach for chili germplasm Differentiation and leaf disease classification

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ABSTRACT

The primary objective of this study was to evaluate the reliability of chili leaf image classification as a method for distinguishing chili germplasm accessions, thereby supporting germplasm conservation efforts. Traditional methods for identifying chili varieties rely on manual leaf observation, which is labor-intensive and error-prone, highlighting the need for advanced fine-grained classification techniques. To address this challenge, we introduce a curated chili leaf database, JNUCLS, to support future research. We evaluate state-of-the-art deep learning methods for chili variety classification and propose a novel approach, LGENetB4CA, which combines a modified EfficientNetB4 model with a LeafGabor filter. The EfficientNetB4 incorporates a Coordinate Attention block following its layers to effectively capture both spatial and channel-wise information. This enables the model to focus on critical regions, such as fine-grained leaf textures, while maintaining a global context. The LeafGabor filter enhances intricate leaf details, such as vein structures, while suppressing noisesuch as blurred shadows, significantly improving input quality. Experiments on the JNUCLS and COLD chili datasets demonstrate high accuracy in chili variety and leaf disease classification, with LGENetB4CA achieving 89.61% accuracy on JNUCLS and 85.90% on COLD chili. These findings highlight the potential of leaf image classification as an effective, cost-efficient tool for exploring phenotypic diversity among chili cultivars. The proposed method also demonstrates promise for broader applications, including plant classification systems, targeted crop management, agricultural product tracking, market analysis, and biodiversity preservation.

1. Introduction

Artificial intelligence (AI) has significantly transformed smart agriculture by providing innovative solutions across various domains (Dhanya et al., 2022). Through deep learning-based computer vision, AI facilitates the precise automation of critical agricultural tasks, such as assessing seed quality (Loddo et al., 2021, Loddo and Di Ruberto, 2021, Zhang et al., 2022), predicting crop yields (Zhang et al., 2023), automating harvesting (Lijia et al., 2023), controlling weeds (Espejo-Garcia et al., 2020), detecting plant diseases (Maurya et al., 2024, Yang et al., 2022), and identifying plant species for applications in agronomy, conservation, and natural product development (Grinblat et al., 2016; Sau et al., 2018; Sau et al., 2019; Loddo et al., 2021; Oppong et al., 2022 Sep). This technology addresses the limitations of traditional manual

methods, which are time-consuming, costly, and prone to errors. Chili peppers, an agricultural product of considerable global economic value, are widely used as a spice, in chili sauces, and as a food ingredient (Hespeler and HamidrezaNemati, 2021). As with other crops, deep learning-based computer vision technologies have been developed to tackle challenges such as crop detection, fruit quality classification, disease detection, and species identification.

In the domain of crop detection for automated harvesting, Steven et al. (Hespeler and HamidrezaNemati, 2021) compared Mask R-Convolutional Neural Network (CNN) and YOLOv3 using RGB and thermal images of chili peppers in varied environments. YOLOv3 achieved amAP of 0.97 and was over ten times faster than Mask R-CNN, making it suitable for real-time robotic harvesting. Shapiee et al. (Shapiee et al., 2022) evaluated YOLO V4 Darknet, YOLO V4 Tiny, Faster R-CNN, and

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Table 1
Summary of chili classification.

Research study	Years	Problems	Methods	Datasets
Steven et al. (Hespeler and HamidrezaNemati, 2021)	2021	Crop detection	Mask R-CNN and YOLOv3	RGB and thermal images of chili peppers in an environment of various amounts of debris
Shapiee et al. (Shapiee et al., 2022)	2022	Crop detection	YOLO V4 Darknet	298 images of bird's eye chili and 324 images of healthy and unhealthy capsicum leaves
Ram et al. (Ram et al., 2023)	2023	Crop detection	YOLOv5	Ripened red chilies in real-world images, totaling 1078 images
T. Purwaningsih et al. (Purwaningsih et al., 2018)	2018	Chili quality Classify	CNN	70 images, 2 classes (feasible and unsuitable chili)
Sudianto et al. (Sudianto et al., 2020)	2020	Chili quality Classify	YOLO	100 images, 2 classes (50 quality A chili images and 50 images of quality B chili)
Cruz-Domínguez et al. (Cruz-Domínguez et al., 2021)	2021	Chili quality Classify	Grayscale based Histogram + ANN	850 images, 3 classes, (Three quality levels of the Guajillo chili)
Hendrawan et al. (Hendrawan, 2021)	2021	Fruit quality assessment	GoogLeNet	1200 images, 3 classes (Three maturity levels)
NageswararaoNaik et al. (NageswararaoNaik et al., 2022)	2022	Leaf disease classification	Squeeze-and-excitation-based CNN (SECNN)	5100 images, 6 classes (down curl of a leaf, Geminivirus, Cercospora leaf spot, yellow leaf disease, up curl disease, and healthy leaf)
Bezabih et al. (Bezabih et al., 2023)	2023	Leaf disease classification	Concatenated CNN model (VGG16 and AlexNet)	1596 images, 4 classes (common rust, healthy fruit, fruit rot disease, grey leaves, and healthy leaves)
Krishna Pratap et al. (Krishna Pratap and Suresh Kumar, 2023)	2023	Leaf disease classification	EfficientLeafNetB4	2000 images, 5 classes (Healthy leaf, Up curl, Down curl, Gemini virus, Cercospora leaf spot)
Suwarningsih et al. (Suwarningsih et al., 2022)	2022	Chili varieties classification	DenseNet-121	1114 images, 12 classes of chili varieties

EfficientDet for classifying bird's eye chili, with YOLO V4 Darknet achieving the best mAP of 76.54 % and 75.69 % on a dataset of 298 bird's eye chili images and 324 capsicum leaf images. Similarly, Ram et al. (Ram et al., 2023) developed a red chili detection system using YOLOv5 on a customized dataset of 1078 images, achieving 95 % accuracy by enhancing image quality through preprocessing.

In fruit quality classification, Purwaningsih et al. (Purwaningsih et al., 2018) used a CNN model to classify chili pepper images as either feasible or unsuitable. Feasible chili refers to peppers with a fresh red color and no visible damage, while unsuitable chili exhibits a blackish-red hue with signs of damage or rot. The CNN model achieved a classification accuracy of 80 % on test data using RGB images. Sudianto et al. (Sudianto et al., 2020) developed a deep learning-based decision support system to classify chili quality into two categories: (1) A –quality chili (whole, clean, healthy, normally shaped, curly red chili without defects) and (2) B-quality chili (smaller, curly red chili, less than 10 cm in length, with abnormal shape or damage). Cruz-Domínguez et al. (Cruz-Domínguez et al., 2021) created a classification system for dried chili using artificial neural networks to recognize peppers based on size and color. This system, which employed 8-bit grayscale image histograms, achieved an accuracy of 82.13 % when tested on three quality levels of Guajillo chili. Hendrawan et al. (Hendrawan, 2021) utilized a CNN to classify the maturity stages of large green chili into three levels (maturity 1 at 34 days after anthesis (DAA), maturity 2 at 47 DAA, and maturity 3 at 60 DAA) using deep learning and computer vision.

In chili leaf disease classification, Nageswararao Naik et al. (NageswararaoNaik et al., 2022) utilized 12pre-trained deep learning networks, including AlexNet, DarkNet53, DenseNet201, EfficientNetb0, InceptionV3, MobileNetV2 (Sandler et al., 2018), NasNetLarge, ResNet101, ShuffleNet, SqueezeNet, VGG19 (Simonyan and Zisserman, 2014), and XceptionNet. They used a dataset containing five main chili leaf diseases—down curl, Geminivirus, Cercospora leaf spot, yellow leaf disease, and up curl—captured with a digital camera and labeled accordingly. VGG19 achieved the highest accuracy without augmentation (83.54 %), while DarkNet53 performed best with augmentation (98.82 %). To further enhance performance, the authors introduced a squeeze-and-excitation-based CNN (SECNN) model, which achieved accuracies of 98.63 % without augmentation and 99.12 % with augmentation. The SECNN model also demonstrated a 99.28 % accuracy on the PlantVillage dataset, classifying 43 different leaf classes. Bezabih et al. (Bezabih et al., 2023) proposed a combined neural network model that integrates features from VGG16 and AlexNet for pepper disease classification using fully connected layers. This model, after image

preprocessing, noise removal, segmentation, feature extraction, and classification, achieved a training accuracy of 100 %, a validation accuracy of 97.29 %, and a testing accuracy of 95.82 %. Krishna Pratap et al. (Krishna Pratap and Suresh Kumar, 2023) highlighted the effectiveness of deep learning architectures in detecting leaf diseases, particularly the fine-tuned EfficientLeafNetB4, which outperformed ResNet-50, DenseNet-121, MobileNet-V2, and VGG-16 in chili leaf disease detection.

Chili pepper variety classification is vital for evaluating genetic diversity, enhancing genetic resources, optimizing yield, adapting to soil conditions, improving pest resistance, and meeting market preferences. Raihana et al. (Ridzuan et al., 2018) investigated the genetic diversity of 14 chili pepper genotypes based on morphological, physiological, and yield traits under a fertigation system in soilless media over two planting seasons to estimate genetic variance components, heritability, and genetic advance for future hybridization. Lucienne et al. (Waongo et al., 2021) analyzed 59 accessions using a Randomized Complete Block Design with three repetitions, evaluating 12 qualitative and 12 quantitative variables. Their study revealed substantial diversity among the accessions, with all qualitative variables except seed color effectively discriminating between them, and found significant correlations, particularly between leaf and fruit traits, supporting future chili pepper selection. Suwarningsih et al. (Suwarningsih et al., 2022) noted that automated cellular and molecular methods for cultivar identification are expensive and energy-intensive, lacking enzyme support for molecular validation compared to manual leaf identification. To address this, they used deep CNN models—AlexNet, VGG16, Inception-v3, and DenseNet-121—to classify chili varieties based on 10-day-old leaf images. With a preprocessing strategy to enhance the dataset and improve classification accuracy, their study achieved accuracies between 70.18 % and 78.37 %, with DenseNet-121 performing best at 78.37 %, demonstrating robust performance despite the small dataset.

Based on the survey results presented in Table 1, deep learning approaches (GoogLeNet, DenseNet-121, EfficientNetB4)combined with Gaussian filter-based feature preprocessing (Krishna Pratap and Suresh Kumar, 2023) have proven effective in classifying chili plants based on leaf features, particularly in identifying diseased leaves and evaluating chili quality. However, limited methods (AlexNet, VGG16, Inception-v3, and DenseNet-121) have been investigated for classifying chili varieties. The current number of chili varieties considered is relatively small (12 chili varieties), with an achieved accuracy of 78.37 % (Suwarningsih et al., 2022). This highlights challenges, such as the need to classify a more extensive range of varieties and the development of more efficient



Fig. 1. The image of an experimental garden (a) and the capture device (b).

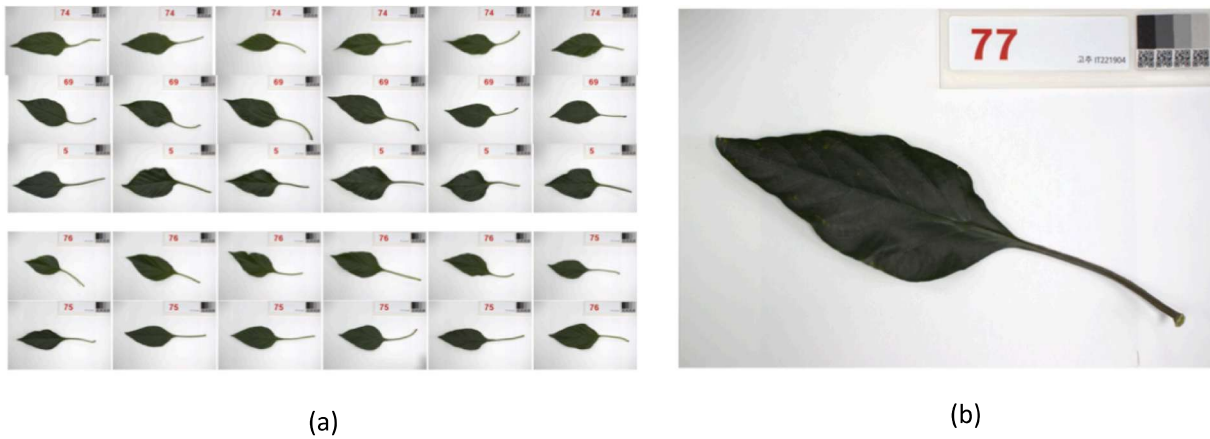


Fig. 2. Some examples of JNU-CLS database: (a) six samples of chili varieties: 74, 69, 5, 76, and 75, respectively, (b) a sample of chili varieties.

deep learning methods for chili varieties classification. In addition, numerous methods for plant species identification based on leaf characteristics have been published, using either hand-crafted feature-based approaches (EnesYigit and AbdurrahimToktas, 2019) or deep learning (Lee et al., 2017). Hand-crafted methods, such as those by Chi et al. (Zheru et al., 2003), Cope et al. (Cope et al., 2010), and Chaki et al. (Chaki et al., 2015), used Gabor filters and texture analysis, while Zhimin Lv et al. (ZhiminLv, 2023) combined LBP and HOG features with PCA for leaf recognition. However, these methods suffer from poor generalization and low efficiency. In contrast, deep learning has revolutionized plant classification, with CNNs and deep residual networks such as ResNet26 proving highly effective. Recent advances include Roopashree et al.'s (Roopashree and Anitha, 2021) evaluation of the DeepHerb dataset with various CNN architectures, Conde et al.'s (Conde et al., 2106) ViT framework, and Van Hieu et al.'s (Van Hieu et al., 2023) PlantKViT, which combines ViT with KNN for forest plant identification.

This research focuses on analyzing the diversity of chili varieties by evaluating deep learning-based classification methods and proposing an efficient algorithm that combines a modified EfficientNetB4 model with Gabor filter-based preprocessing. The contributions of this paper are as follows:

- Introduction of a chili leaf database cultivated and collected by experts, designed for precise cultivar characterization.
- Evaluation of phenotypic diversity in chili varieties using state-of-the-art deeplearning methods for leaf classification.
- Proposal of the LGNetB4CA method, combining a modified

EfficientNetB4 model with a Coordinate Attention block and a Leaf-Gabor filter.

– Evaluation of deep learning models (ViT, DenseNet121, InceptionV3, EfficientNetB3, B4, and LGNetB4CA) demonstrates that LGNetB4CA outperforms the others in chilivarieties and leaf disease classification.

The remainder of this paper is structured as follows: Section 2 describes the dataset, Section 3 elaborates on the proposed method, Section 4 showcases the experimental results, and Section 5 concludes the paper.

2. The JNUCLS dataset

To evaluate chili variety diversity for conservation and genetic resource enhancement, the JNUCLS database includes labeled leaf images of chili varieties collected and cultivated at the Rural Development Administration (RDA) in Jeonju, Republic of Korea. The camera setup (Fig. 1) followed methodologies outlined by Yu et al. (Yu et al., 2023) (Yu et al., 2023) and other leaf varieties classification studies (Diago et al., 2013, Karadeniz et al., 2024). The indoor studio, measuring 800 × 800 × 800 mm, was equipped with an 18 W white LED light (5600 K, CN-T96, Republic of Korea) to minimize ambient light interference and ensure consistent illumination, thereby preventing data distortion from shadowing effects. Images were captured using a Canon EOS D200II camera (EF-S 18–55 mm lens, 24.1-megapixel CMOS sensor). Initial settings (ISO 200, 35 mm focal length, 1/25 exposure) were adjusted to

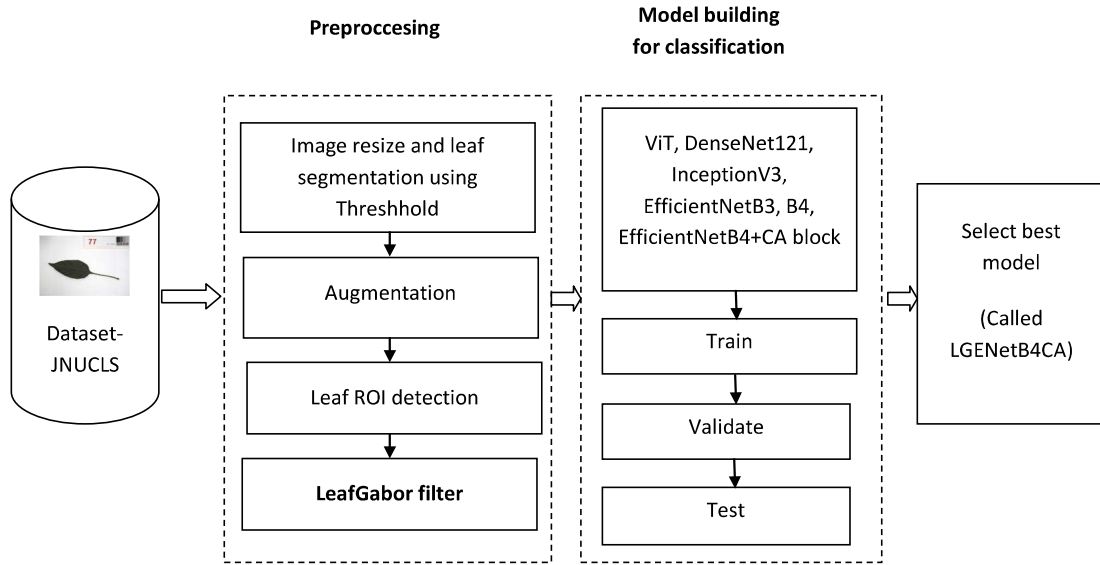


Fig. 3. The proposed framework for leaf-based chili varieties classification.

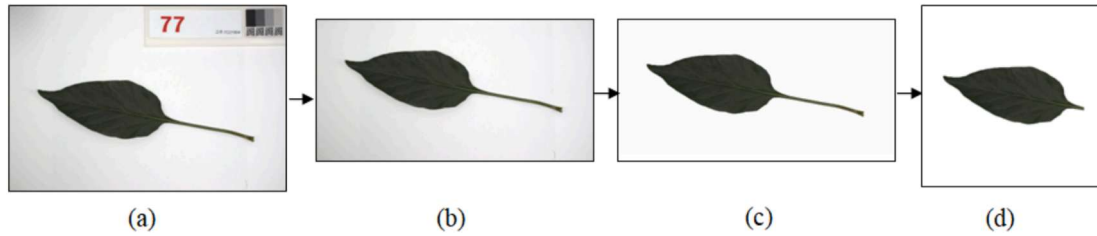


Fig. 4. The processing steps for extracting leaf ROI images: (a) original image, (b) leaf image without a label, (c) leaf image without background, processed with a grayscale threshold of 200, and (d) leaf image without leaf stalk, extracted using YOLOv8 ((Leaf Detection Computer Vision Project, 2023)).

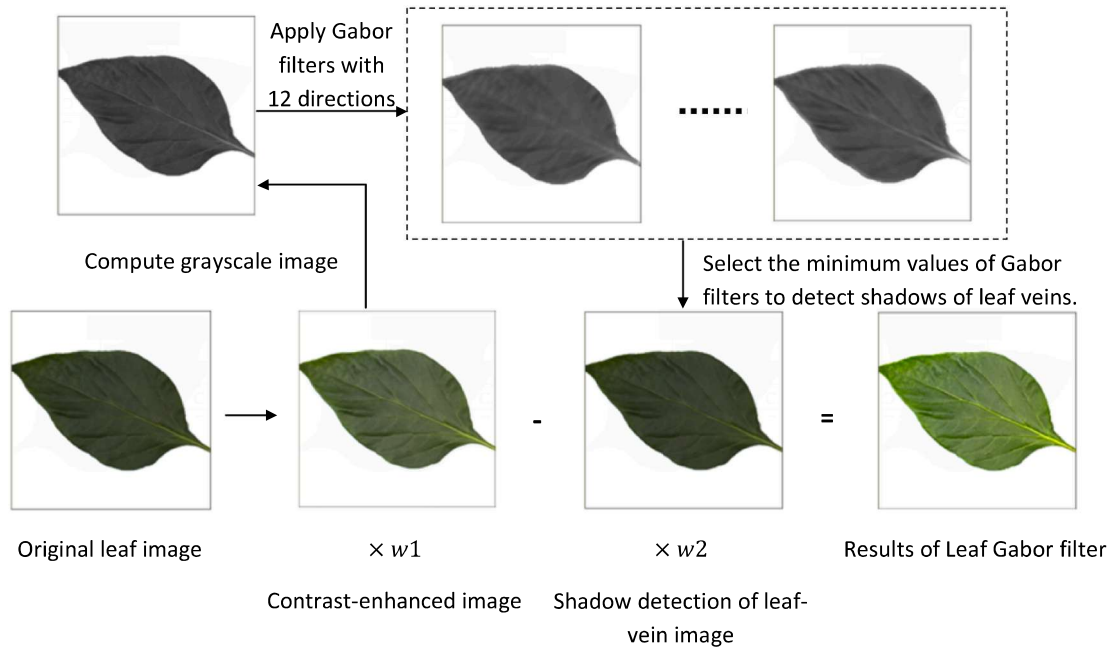


Fig. 5. The processing flowchart of the proposed LeafGabor filter.

ISO 100, 18 mm focal length, and 1/13 exposure to correct lens-induced distortion using the camera's built-in correction function. To ensure consistent leaf positioning, custom 3D panels with crop name tags, four

10 mm grayscale markers, and QR codes were included in the setup. Chili varieties were cultivated under controlled conditions, including regulated temperature, irrigation, and fertilization. The experiment

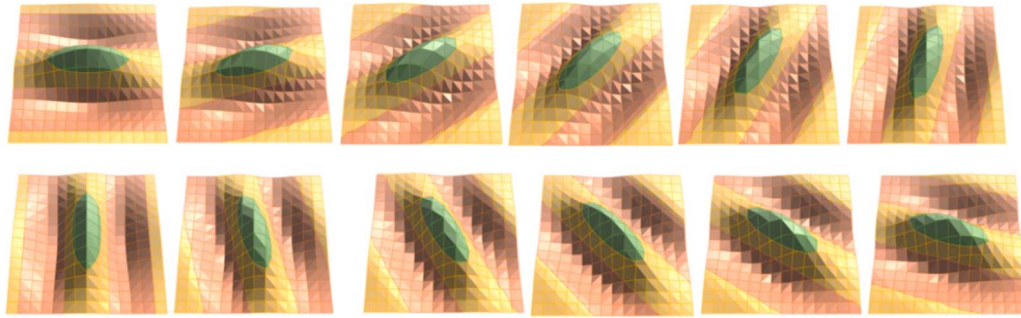


Fig. 6. 15×15 Gabor filters at the directions of $\theta_j = \pi(j - 1)/2, j = 1, 2, \dots, 12$, respectively.

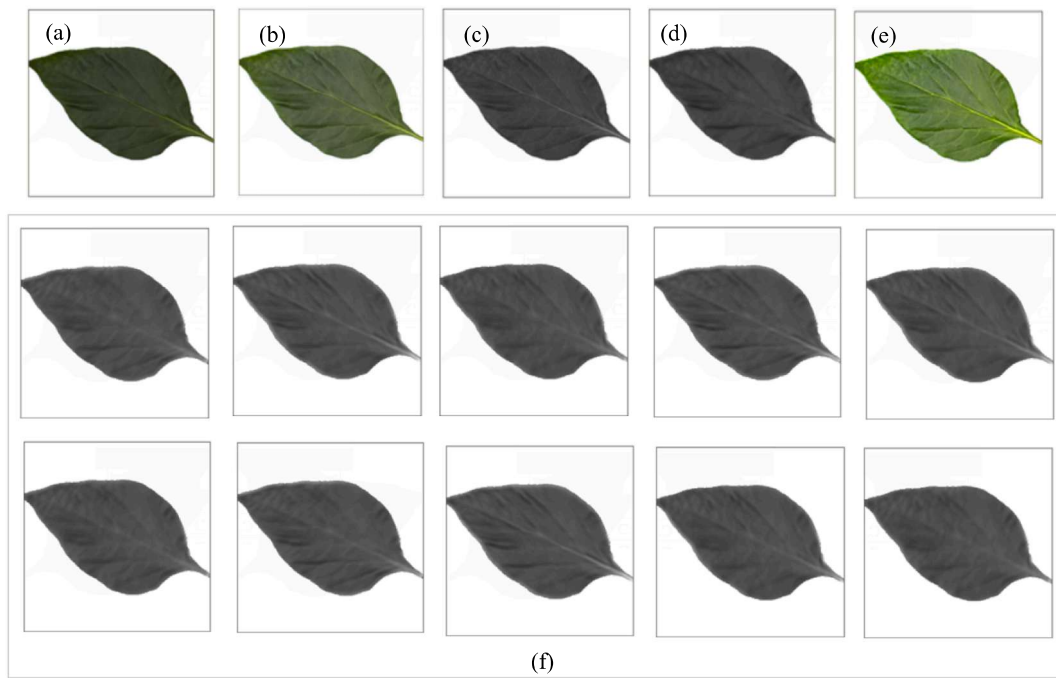


Fig. 7. Original leaf ROI image (a), contrast-enhanced image (b), Grayscale image (c), Minimum Response of Gabor filters (d), and Resulting image of LeafGabor filters (e). There are 12 resulting images of Gabor filters at directions $\theta_j = (\pi(j-1))/2, j = 1, 2, \dots, 12$ with $\gamma = 1.4, \mu = 16, \sigma = 0.25, w1 = 2, w2 = 1.5$, respectively (f).

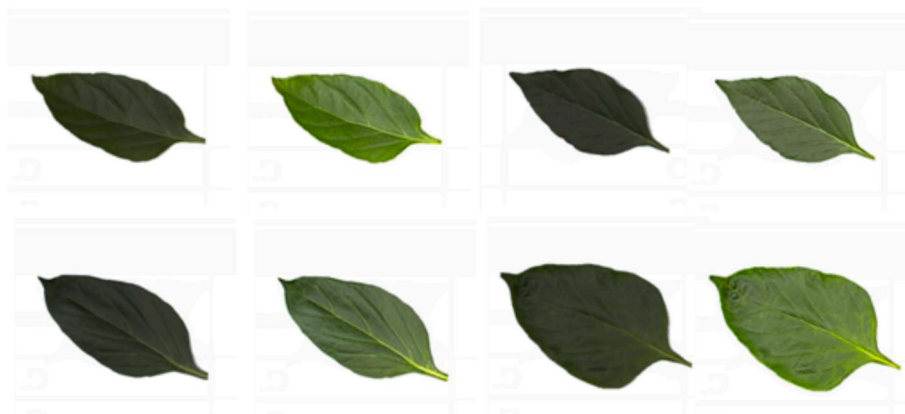


Fig. 8. Some Samples of Image Quality Enhancement using the LeafGabor filter in JNUCLS Dataset.

involved 77 chili varieties, each randomly arranged in greenhouse plots and identified by species codes. For each variety, six plants were grown per plot, and six leaf samples were collected. Sampling was conducted at the flowering stage, with leaves selected based on standardized criteria

such as area, weight, and color. Images were captured with uniform lighting, a consistent distance, and a fixed shooting angle using a Canon digital camera. All images were saved in JPG format with a resolution of 6024×4024 pixels (Fig. 2). The database initially contained 462 images

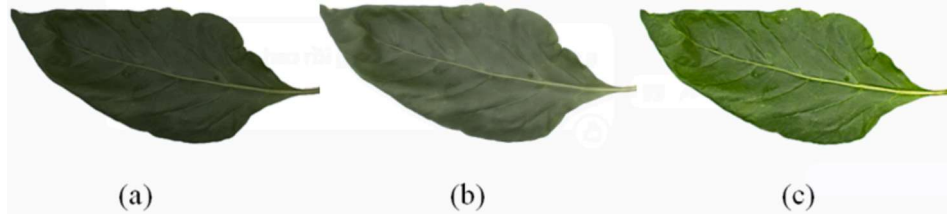


Fig. 9. (a) Original image, (b) CLAHE image (Krishna Pratap and Suresh Kumar, 2023), (c) LeafGabor-filtered image.

Table 2
LGENetB4CA for Leaf-Based Classification of 77 Chili Species.

The modified model: LGENetB4CA	
Layer type	Output shape
Input (LeafGabor images)	$300 \times 300 \times 3$
EfficientNetB4 Base Model	$10 \times 10 \times 1792$
Coordination Attention (CA)	$10 \times 10 \times 1792$
Global Average Pooling	1792
Batch Normalization	1792
Dense (256 units, ReLU)	256
Dropout	256
Dense (Output classes, Softmax)	77

prior to augmentation.

3. Proposed methods

This study evaluates the reliability of leaf image classification for distinguishing chili germplasm accessions using the fine-grained dataset presented in Section 2. Initially, state-of-the-art deep learning methods for chili variety classification are explored. Atila et al. (ÜmitAtila et al., 2021) demonstrated the superior accuracy of the EfficientNetB4 model for classifying diseased leaves compared to other state-of-the-art CNN methods. Additionally, Hou et al. (Hou et al., 2021) proposed the Coordinate Attention mechanism, which is initially comparable to the (Squeeze-and-Excitation)SE attention module in EfficientNet yet differs in its ability to embed spatial positional information into channel attention maps, thereby enhancing feature aggregation. Building on these findings, this study introduces a novel model by integrating a Coordinate Attention (CA) block into EfficientNetB4, enabling it to focus on critical image regions, such as fine-grained leaf textures, while preserving global context. Additionally, Pratap et al. (Krishna Pratap and Suresh Kumar, 2023) improved EfficientNetB4's performance in chili

leaf disease recognition by applying preprocessing techniques such as Histogram Equalization and Gaussian filters. In parallel, the Gabor filter has demonstrated high effectiveness in texture analysis, edge detection, and feature extraction (Zheru et al., 2003; Le and Van, 2012). Leveraging these strengths, the LeafGabor filter is introduced to capture intricate leaf details, enhance vein clarity, and eliminate blurred shadows, enabling EfficientNetB4 to maximize the benefits of deep learning techniques. As a result, this study presents a novel fine-grained classification method, LGENetB4CA, which combines EfficientNetB4 with a CA block and a LeafGabor filter to achieve enhanced classification performance. A visual representation of the proposed framework for chili variety classification is illustrated in Fig. 3. Detailed descriptions of each processing step within the method are provided in the subsequent subsections.

3.1. Leaf Region of interest (ROI) segmentation

Detecting the leaf area, referred to as ROI image segmentation, is an important step in the feature extraction stage. Initially, the images were resized to 600×400 pixels, followed by applying a threshold to isolate the leaf area and remove the background. The YOLOv8 method (Shapiee et al., 2022, Ram et al., 2023); (Leaf Detection Computer Vision Project, 2023) was then used for leaf detection, placing the detected leaves at the center of the image with a size of 400×400 pixels. Fig. 4 illustrates this process of extracting ROI images.

3.2. Image augmentation

Data augmentation involves introducing diverse examples into the training dataset, thereby enhancing the efficacy and results of deep learning algorithms. Augmented training data improves both the performance and robustness of models (Krishna Pratap and Suresh Kumar, 2023; Bello, et al., 2021). By simulating various conditions during

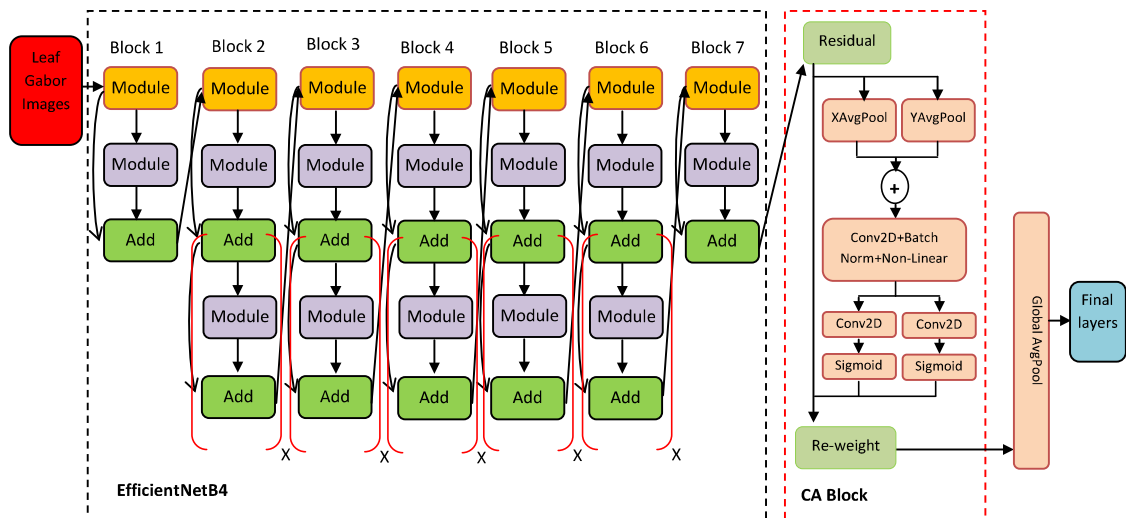


Fig. 10. Architecture of our proposed method-LGENetB4CA.

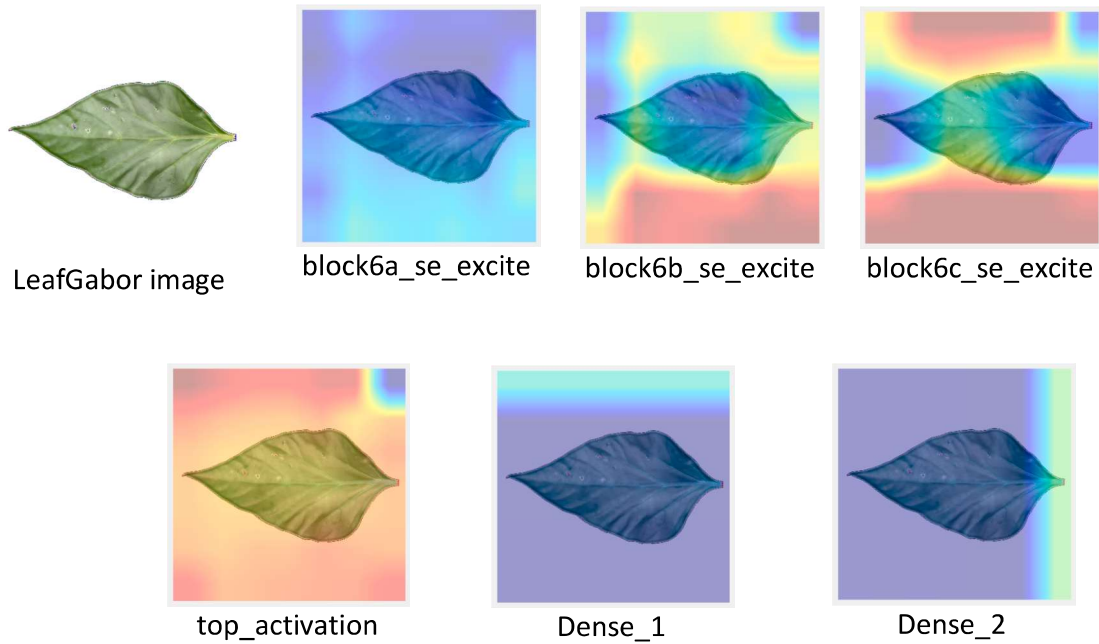


Fig. 11. Visualizations of some layers of our proposed method using GradCAM, where the Dense_1 and Dense_2 layers belong to the CA block.

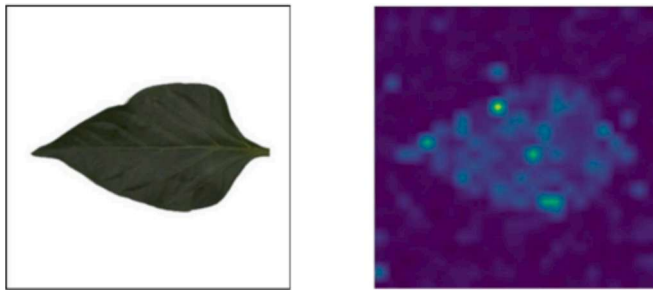


Fig. 12. The original image (a) and its attention map from the last layer of ViT (b).

training, the model's ability to handle unseen examples is strengthened. The quantity of augmented images generated during data augmentation depends on techniques such as flipping, rotation, width shift, height shift, and zooming, along with their respective parameters. In the current experiments, we configured the following parameters: a rotation range of 20 degrees, a width shift range of 0.1, a height shift range of 0.1, and a zoom range of 0.2 relative to the image size.

3.3. LeafGabor filter for extracting leaf vein features

Leaves are commonly used in plant species classification due to their distinctive features, such as shape, texture, and vein patterns (EnesYigit and AbdurrahimToktas, 2019, Lee et al., 2017). The Gabor filter, a convolution-based filter that combines a Gaussian component with a sinusoidal term, is a highly effective tool commonly applied to enhance ridge patterns or to detect edges and textures in images (Hou et al., 2021, Le and Van, 2012).

Table 3
Parameters of the Datasets.

Dataset	Each class Training	Validation	Testing	All class Training	Validation	Testing
Dataset 1	4	1	1	23100 = 77*4 + 22792	77	77
Dataset 2	2	2	2	23100 = 77*2 + 22946	154	154

To enable EfficientNetB4 to leverage deep learning more effectively, we propose the LeafGabor filter. This filter applies Gabor filters with varying orientations to remove shadows, clarify leaf veins, and capture intricate leaf details, thereby improving image quality. Given image I , the LeafGabor filter consists of three processing steps (Fig. 4):

1 Enhancing image contrast using gamma correction

$$f = I^\gamma$$

where f is the contrast-enhanced image, γ is gamma value.

2 Applying Gabor filters with 12 different orientations at each pixel to detect the direction of leaf veins. Gabor filters are commonly used to detect local lines (Sudianto et al., 2020, Cruz-Domínguez et al., 2021) and employed here to detect shadows of leaf veins as follows

$$f_g = \min_{\theta_i = \frac{\pi(i-1)}{12}, i=1,2,\dots,12} f * G(x, y, \theta_i, \mu, \sigma) \quad (2)$$

Table 4
The results of deep learning models on dataset 1 (%).

Methods	Precision	Recall	F1-Score	Accuracy
ViT	82.47	87.01	83.98	87.01
LGViT	86.36	90.91	87.88	90.91
DenseNet121	72.29	80.52	74.89	80.52
LGDenseNet121	83.12	88.31	84.85	88.31
InceptionV3	76.62	84.42	79.22	84.42
LGINceptionV3	85.06	89.61	88.58	89.61
EfficientNetB3	89.61	92.21	90.48	92.21
LGEfficientNetB3	92.21	94.81	93.07	94.81
EfficientNetB4	88.31	92.21	89.61	92.21
LGEfficientNetB4	92.86	94.81	93.51	94.81
EfficientNetB4CA	90.26	93.51	91.34	93.51
LGNetB4CA	94.16	96.10	94.81	96.10

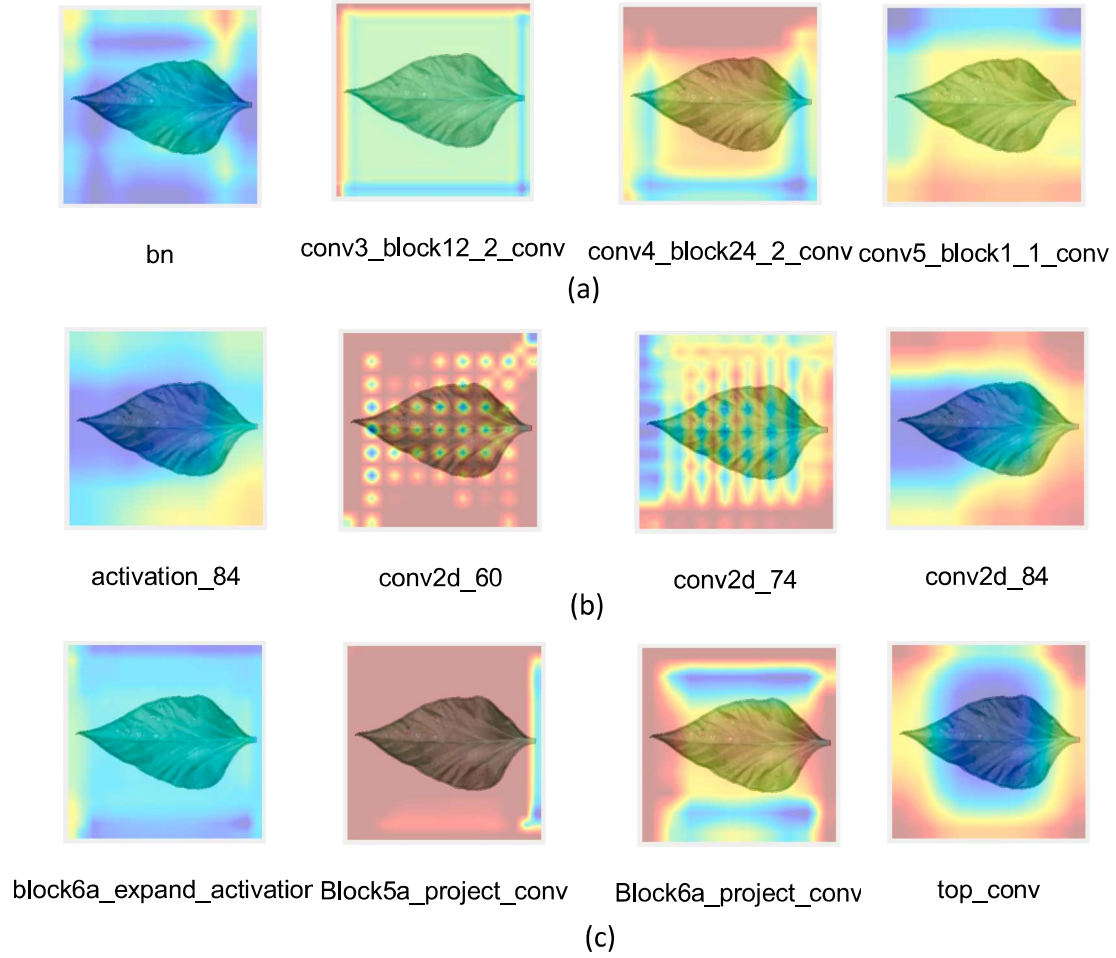


Fig. 13. Feature visualization of some layers from the (a) DenseNet121, (b) InceptionV3, and (c) EfficientNetB3 models using GradCAM, respectively.

where f_g is the leaf vein shadow image obtained by convoluting image f using the real part of the Gabor filter with 12 directions, and the 2D circular Gabor filter has the form

$$G(x, y, \theta_i, \mu, \sigma) = \frac{1}{2\pi\sigma^2} \exp\left\{-\frac{x^2 + y^2}{2\sigma^2}\right\} \exp\{2\pi i(\mu x \cos\theta + \mu y \sin\theta)\} \quad (3)$$

Here, $i = \sqrt{-1}$, μ is the frequency of the sinusoidal wave, θ_i controls the orientation of the function, and σ is the standard deviation of the Gaussian envelope. The real parts of the Gabor filter in the 12 directions are presented in Fig. 5.

3 Adding the resulting vein-filtered image to the contrast-enhanced image yields an image with improved quality

$$f_q = w1 \times f - w2 \times f_g \quad (4)$$

Where f_q represents the resulting image after applying the LeafGabor filter. $w1$ and $w2$ denote the weights determining the contribution of the contrast-enhanced image f and the leaf vein shadow image f_g , respectively.

Fig. 6 illustrates the processing steps for applying the LeafGabor filter: (a) original leaf ROI image, contrast-enhanced image, grayscale image, minimum response of Gabor filters, the resulting image of LeafGabor filters, and 12 resulting images of Gabor filters corresponding to directions $\theta_j = (\pi(j-1))/2$, where j ranges from 1 to 12. In our experiments, parameters were selected as the following: $\gamma = 1.4$, $\mu = 16$, $\sigma = 0.25$, $w1 = 2$, $w2 = 1.5$. Fig. 7 illustrates several enhanced leaf images, where the leaf vein lines are made clearer. Fig. 8 compares the image quality processed by the LeafGabor filter and the CLAHE-based method (Krishna Pratap and Suresh Kumar, 2023). The image processed by the

LeafGabor filter exhibits clearer vein patterns and more distinct texture details compared to the CLAHE-based method.

3.4. LGENetB4CA for leaf-based chili recognition

EfficientNet, proposed by Tan and Le (Tan and Le, 2019), is a renowned family of CNN architectures celebrated for their efficacy in image classification. This family encompasses various variants, denoted from EfficientNetB0 to EfficientNetB7, each characterized by increasing depth, width, and resolution. Notably, EfficientNetB0 represents the smallest and simplest model, while EfficientNetB7 emerges as the largest and most complex variant (Tan and Le, 2019). Among the mid-sized variants within the EfficientNet family, EfficientNetB4 strikes a harmonious balance between model size and performance. In the EfficientNetB4 architecture, SE blocks are utilized alongside MBConv blocks. SE blocks enable the model to learn channel-wise dependencies by enhancing significant channels and suppressing less relevant ones. In addition, Hou et al. (Hou et al., 2021) proposed the CA mechanism. Unlike SE, which transforms a feature tensor into a single feature vector through 2D global pooling, CA factorizes channel attention into two 1D feature encoding processes that aggregate features along the two spatial directions independently. This approach allows long-range dependencies to be captured along one spatial direction while preserving precise positional information along the other. The resulting feature maps are encoded into a pair of direction-aware and position-sensitive attention maps, which are complementarily applied to the input feature map to enhance the representations of the objects of interest. By embedding positional information into the attention mechanism, CA

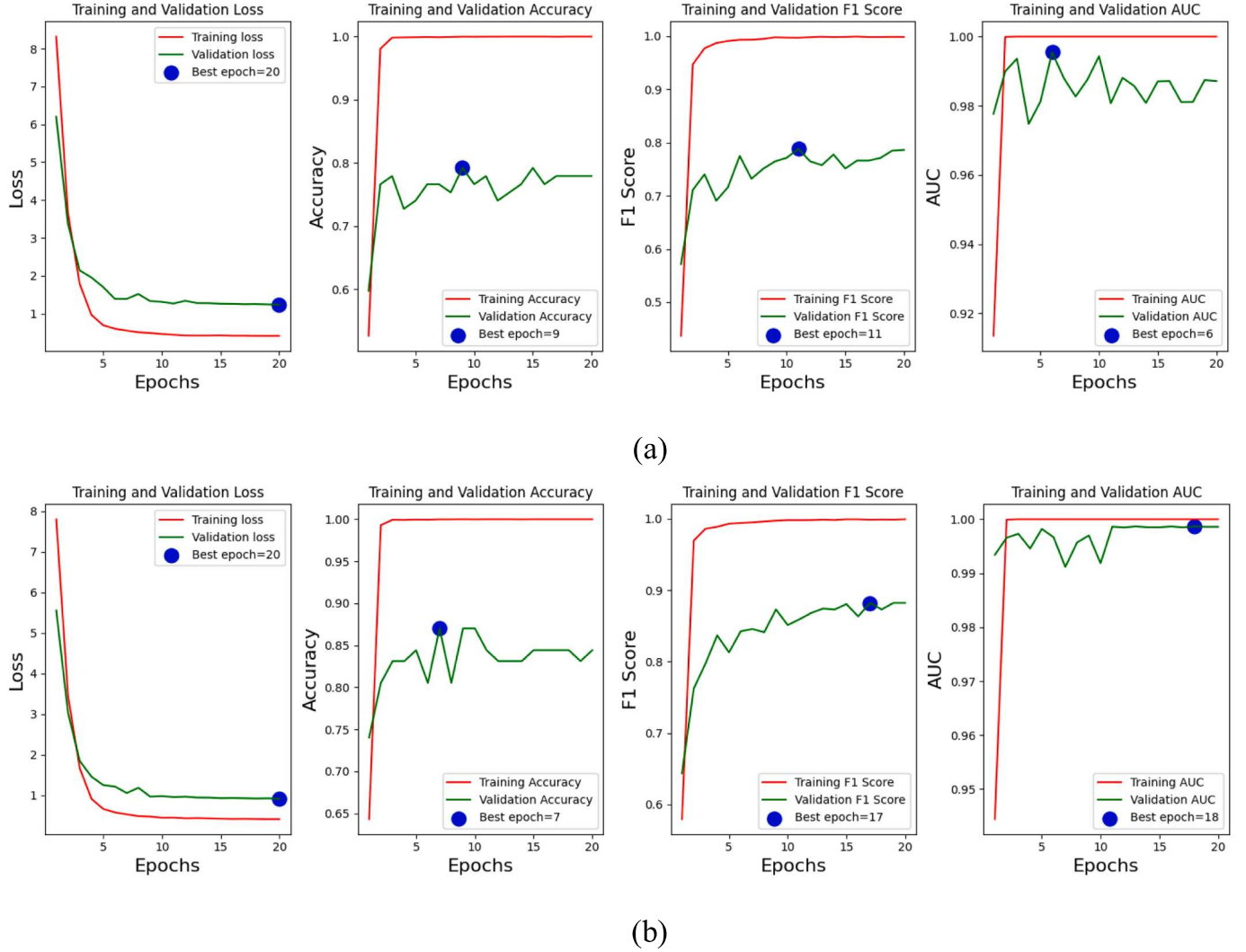


Fig. 14. A comparison chart of methods during training and validation on Dataset 1: (a) EfficientNetB4 and (b) LGENetB4CA, respectively.

improves the model's ability to localize relevant features. This is particularly advantageous for tasks such as chili leaf variety classification, where subtle variations in vein patterns and shapes are critical for accurate identification. With CA, EfficientNetB4, referred to as EfficientNetB4CA, achieves superior refinement of intermediate features, ensuring that the extracted representations are both discriminative and semantically rich. This leads to improved classification accuracy. As a result, this study introduces a novel fine-grained classification method, LGENetB4CA, which integrates EfficientNetB4 with a CA block and a LeafGabor filter to further enhance classification performance. Fig. 9 represents the architecture of LGENetB4CA.

To fully leverage the capabilities of EfficientNetB4, we initialize the model with pre-trained weights from large-scale datasets, such as ImageNet, and fine-tune it on a dataset specifically tailored for chili leaf classification. This adaptation process includes the integration of additional layers: a CA block, a Global Average Pooling layer, and batch normalization. Additionally, custom fully connected layers are added, consisting of a dense layer with ReLU activation, a dropout layer, and a final dense layer with softmax activation. These modifications enhance the model's ability to effectively differentiate between various chili leaf varieties. The model is trained using supervised learning techniques, with detailed specifications provided in Table 2. Fig. 10 illustrates the visual outputs of selected layers from the EfficientNetB4 model and the CA block.

4. Experimental results

Experiments were conducted using the JNUCLS dataset and the public COLD chili dataset (Aishwarya, 2024). The experiments with the JNUCLS dataset aimed to assess the genetic diversity of chili leaves based on their geometric characteristics using deep learning algorithms. In contrast, experiments with the COLD dataset were performed to evaluate the effectiveness of the proposed method in classifying chili leaves collected directly from outdoor chili plants. The effectiveness of the LGENetB4CA algorithm in identifying chili varieties was compared against several state-of-the-art deep learning algorithms, including ViT, DenseNet121, InceptionV3, EfficientNetB3, and EfficientNetB4. In these experiments, we utilized the ViT-B/16 model with ImageNet-21 K pre-trained weights, fine-tuning it on a chili leaf classification dataset (Conde et al., 2106) Waongo et al., 2021; Conde et al., 2106. Fig. 11 illustrates an example of the attention map from the final layer of ViT. The DenseNet121 model (Huang et al., 2017) was adapted for chili leaf classification by using the pre-trained DenseNet121 architecture from ImageNet, enhanced with batch normalization, a dense layer with 256 units, dropout regularization, and a softmax output layer for classifying 77 chili leaf varieties. Similarly, the InceptionV3 model (Szegedy et al., 2016) was modified by replacing the fully connected layers with a global max pooling layer, a dense layer with 256 units, dropout regularization, and a softmax output layer for multi-class classification. The EfficientNetB3 model was applied in a manner similar to EfficientNetB4

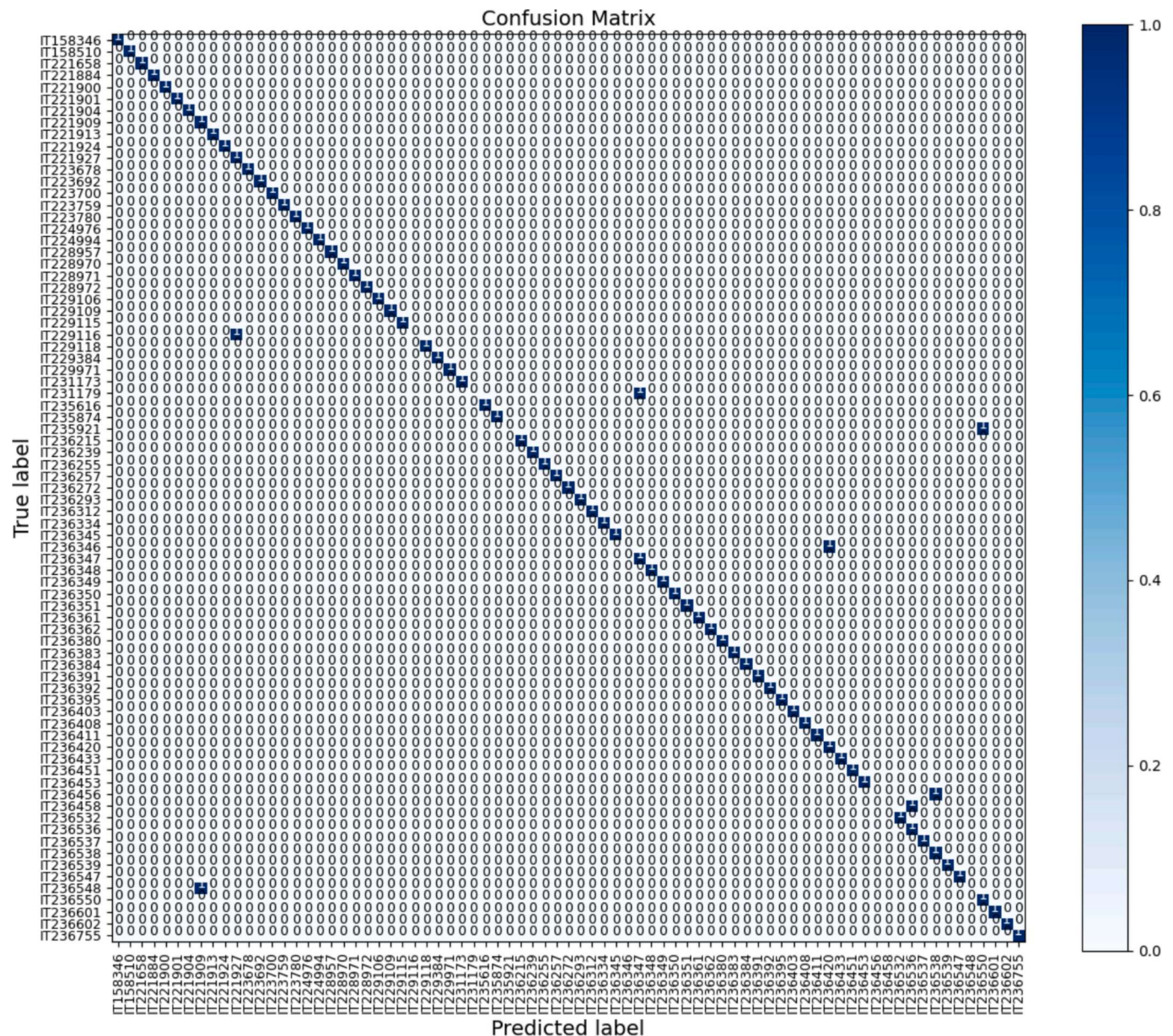


Fig. 15. Confusion Matrix of the result of the LGViT method on dataset1.

Table 5
The results of deep learning models on dataset2 (%).

Methods	Precision	Recall	F1-Score	Accuracy
LGViT	83.39	82.64	81.16	82.46
LGDenseNet121	83.20	81.82	80.01	81.82
LGInception V3	79.65	77.92	76.54	77.92
LGEfficientNetB3	86.06	86.36	84.29	86.36
EfficientNetB4	87.23	85.71	84.29	85.71
CLAEfficientNetB4 (Krishna Prapat and Suresh Kumar, 2023)	87.49	86.36	85.05	86.36
LGEfficientNetB4	89.67	87.66	86.42	87.66
EfficientNetB4CA	88.33	87.01	86.28	87.01
LGNetB4CA	91.13	89.61	88.79	89.61

for chili leaf classification. Feature visualizations of the top_conv layers from the DenseNet121 (Fig. 12a), InceptionV3 (Fig. 12b), and EfficientNetB3 (Fig. 12c) models were generated using GradCAM. The experimental prototype was developed in Python, utilizing Tensor Flow

and other open-source libraries to implement the chili leaf classification models. All models were trained, validated, and tested in the Google Colab Pro environment with an accelerated GPU T4.

4.1. Evaluation metrics

Our proposed method is evaluated using various metrics: Accuracy (Acc), precision (Pre), Recall (Re), and F1-score (F1). These metrics provide a comprehensive assessment of the model's performance across different aspects of classification. Accuracy measures the percentage of correct predictions, while precision estimates the probability of correct positive classifications. Specificity evaluates the percentage of accurately predicted negative classes, and Recall estimates the percentage of accurately predicted positive classes. The F1score provides a single metric for overall performance, making it valuable in situations involving class imbalance. The calculation of these metrics for multi-class classification using macro-averaging is explained by the following Equations (5)–(8) (Sokolova and Lapalme, 2009):

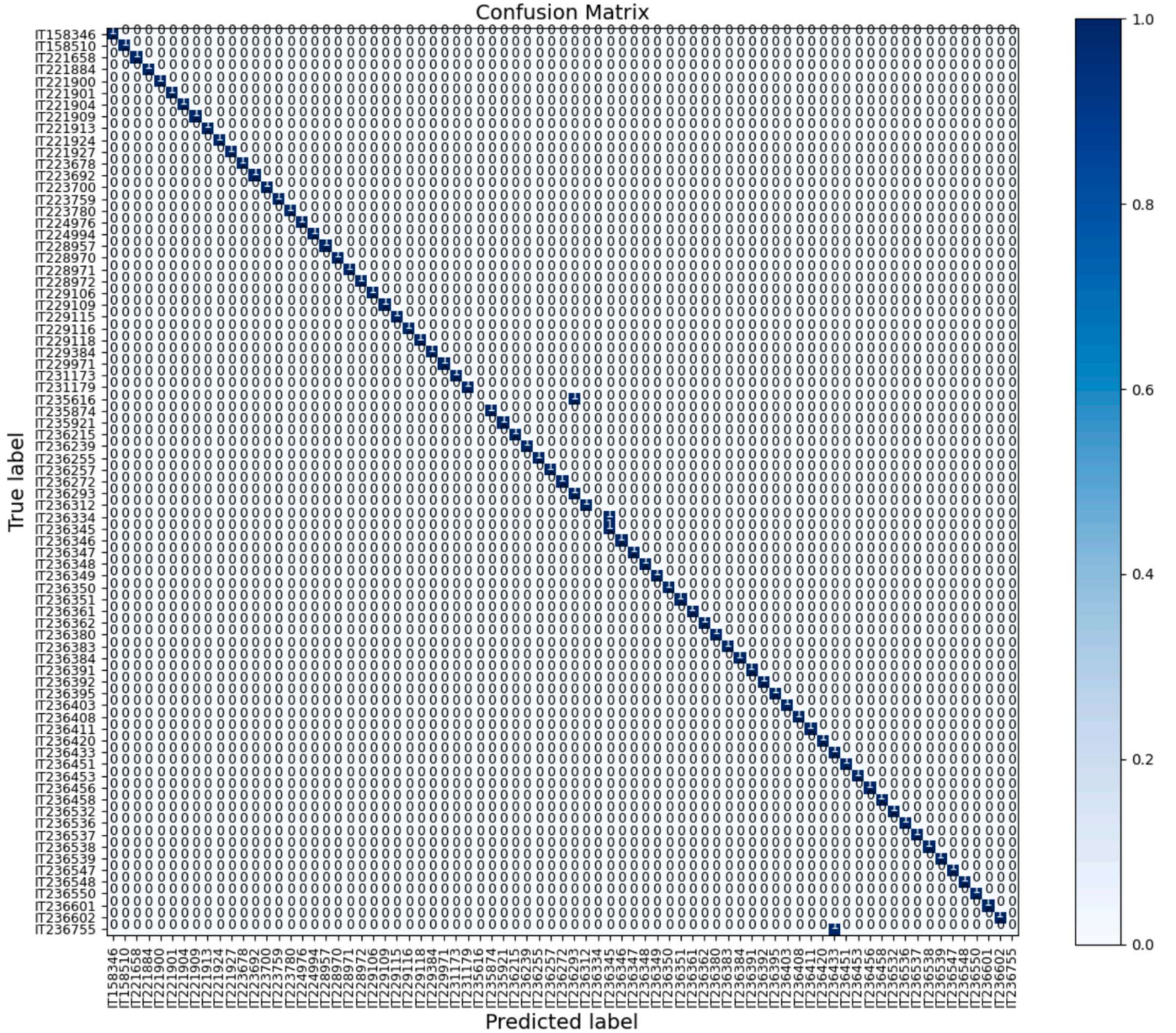


Fig. 16. Confusion Matrix of the result of the LGNetB4CA method on dataset1.

$$AvgAcc = \frac{1}{classes} \sum_i Acc(i), Acc(i) = \frac{TP(i) + TN(i)}{TP(i) + TN(i) + FP(i) + FN(i)}, \quad (5)$$

$$AvgRe = \frac{1}{classes} \sum_i Re(i), Re(i) = \frac{TP(i)}{FN(i) + TP(i)}, \quad (6)$$

$$AvgPre = \frac{1}{classes} \sum_i Pre(i), Pre(i) = \frac{TP(i)}{TP(i) + FP(i)}, \quad (7)$$

$$AvgF_1 = \frac{1}{classes} \sum_i F_1(i), F_1(i) = 2 \times \frac{Pre(i) \times Re(i)}{Pre(i) + Re(i)}, \quad (8)$$

where for class i , TP represents True Positive, TN represents True Negative, FP represents False Positive, and FN represents False Negative.

4.2. Result of experiment 1

In this experiment, we used the JNUCLS dataset, as described in

Section 2, to evaluate the effectiveness of deep learning algorithms in classifying chili varieties. The original dataset was randomly divided into training, validation, and testing groups, and the training set was augmented with 300 images per variety, as detailed in Section 3.2. We established two datasets for the experiment: (1) dataset 1 consisted of 23,100 training images (including 308 original and 22,792 augmented images), 77 validation images, and 77 testing images. dataset 2 consisted of 23,100 training images (including 308 original and 22,946 augmented images), 144 validation images, and 144 testing images. Table 3 presents the parameters of the experimental datasets. The models were trained for 20 epochs with a batch size of 30.

Table 4 presents the experimental results on Dataset 1 for various deep learning methods and the proposed method. Fig. 13 compares the training and validation results on Dataset 1 between the EfficientNetB4 method and the LGNetB4CA method. Figs. 14 and 15 illustrate the confusion matrices of the classification results for the LGViT method and the proposed method on Dataset 1, respectively. The results indicate that EfficientNetB4CA achieves higher accuracy compared to other deep learning methods. When the LeafGabor filter is integrated with these

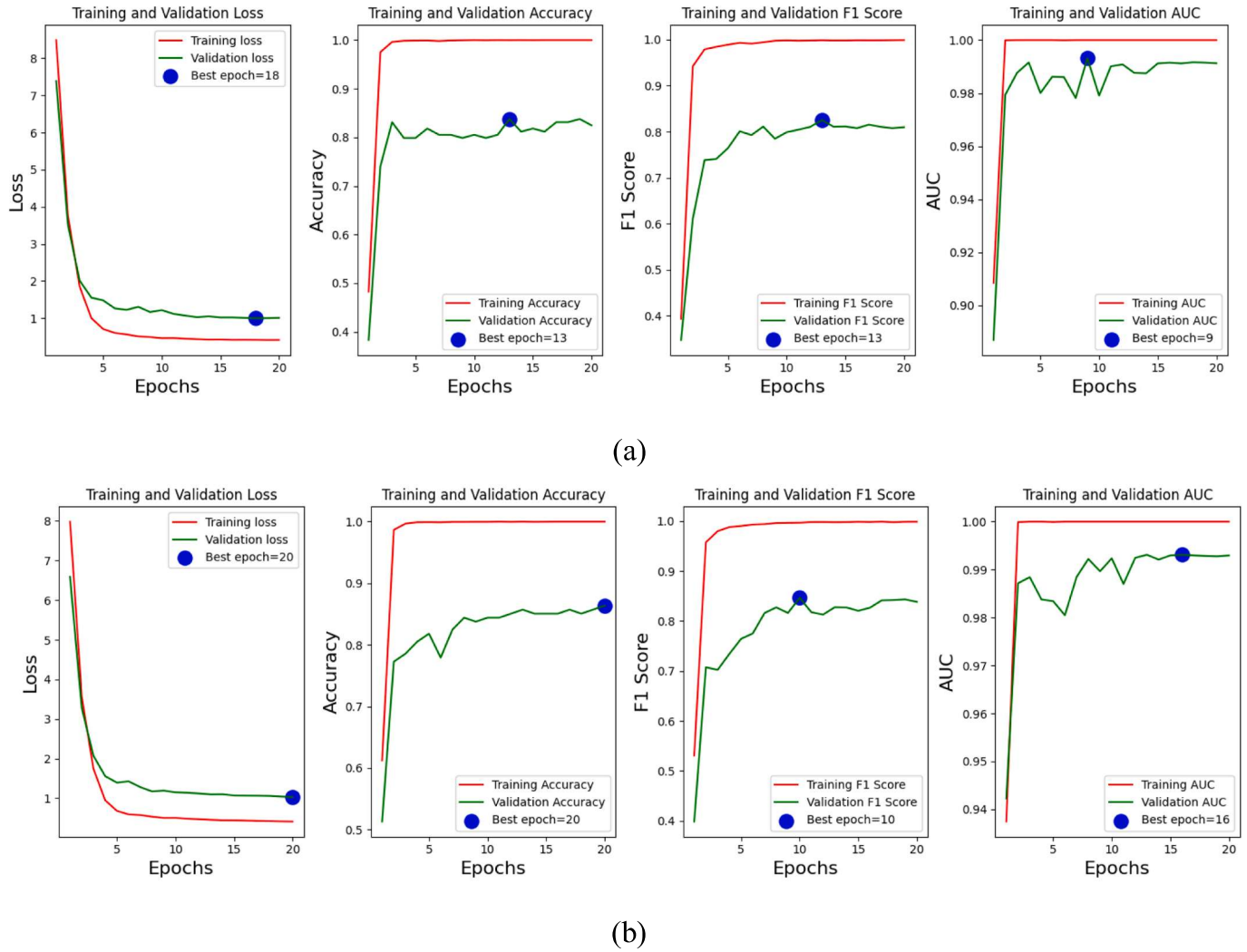


Fig. 17. A Comparison Chart of Various Methods in Training and Validation on Dataset1: (a) EfficientNetB4 and (b) LGE-NetB4CA, respectively.

methods, all approaches show further accuracy improvements across the datasets, with LGE-NetB4CA consistently achieving the highest accuracy. These findings demonstrate that classifying chili varieties based on leaf features yields high accuracy, effectively highlighting the geometric diversity of chili leaf characteristics. Furthermore, the proposed LGE-NetB4CA method proves to be the most effective approach for classifying chili varieties using leaf images.

Table 5 presents the experimental results comparing various methods, including ViT, DenseNet, InceptionNet121, EfficientNetB3, EfficientNetB4, EfficientNetCA with Gabor preprocessing, and EfficientNetB4 with CLAHE preprocessing (Krishna Pratap and Suresh Kumar, 2023) (CLAHEEfficientNetB4). Fig. 16 compares the training and validation results on dataset 2 between EfficientNetB4 and LGE-NetB4CA. Figs. 17 and 18 illustrate the confusion matrices of the classification results for EfficientNetB4 and the proposed method on dataset 2, respectively. The CLAHEEfficientNetB4 method achieves a higher accuracy (86.36 %) compared to EfficientNetB4 (85.71 %) and a lower accuracy compared to LGEfficientNetB4 (87.66 %). This demonstrates that the LeafGabor filter enhances the distinguishing features of leaves more effectively than the CLAHE method, enabling EfficientNetB4 to learn more efficiently. The proposed method achieves the highest recognition accuracy (89.61 %), outperforming all other methods. This further confirms the advantages of the proposed approach and highlights the geometric diversity of chili leaves in variety identification.

4.3. Result of experiment 2

Experiment 2 utilized the COLD chili dataset (Aishwarya, 2024) to evaluate the performance of the proposed method in classifying chili leaves collected directly from outdoor chili plants. The dataset was gathered during the Kharif season using a Canon Mark II D digital camera, capturing images from rural landscapes. The chili leaf images were divided into five categories based on their condition: (1) 2,198 healthy leaves, (2) 2,219 leaves affected by *Cercospora* disease, (3) 2,507 leaves infested with mites and thrips, (4) 2,032 leaves showing signs of nutritional deficiency, and (5) 2,031 leaves with powdery mildew disease. In total, 10,987 images were collected, preprocessed, cropped, and resized before being organized into their respective categories. A sample of images from the COLD chili dataset is shown in Fig. 19. The dataset was split into training and testing sets, where 80 % of the data was allocated for training and 20 % for validation. All models were trained for 40 epochs using a batch size of 30.

Table 6 presents the experimental results of deep learning models applied to the COLD chili dataset. Fig. 20 compares the training and validation performance between EfficientNetB4 and the proposed LGE-NetB4CA method. Figure 21, 22 and 23 display the Confusion Matrices and AUC-ROC curves for the EfficientNetB4CA and LGE-NetB4CA methods, respectively. The EfficientNetB4 method achieved an accuracy of 83.26 %, outperforming DenseNet121 (82.82 %), InceptionV3 (75.53 %), and EfficientNetB3 (82.38 %), though it was slightly

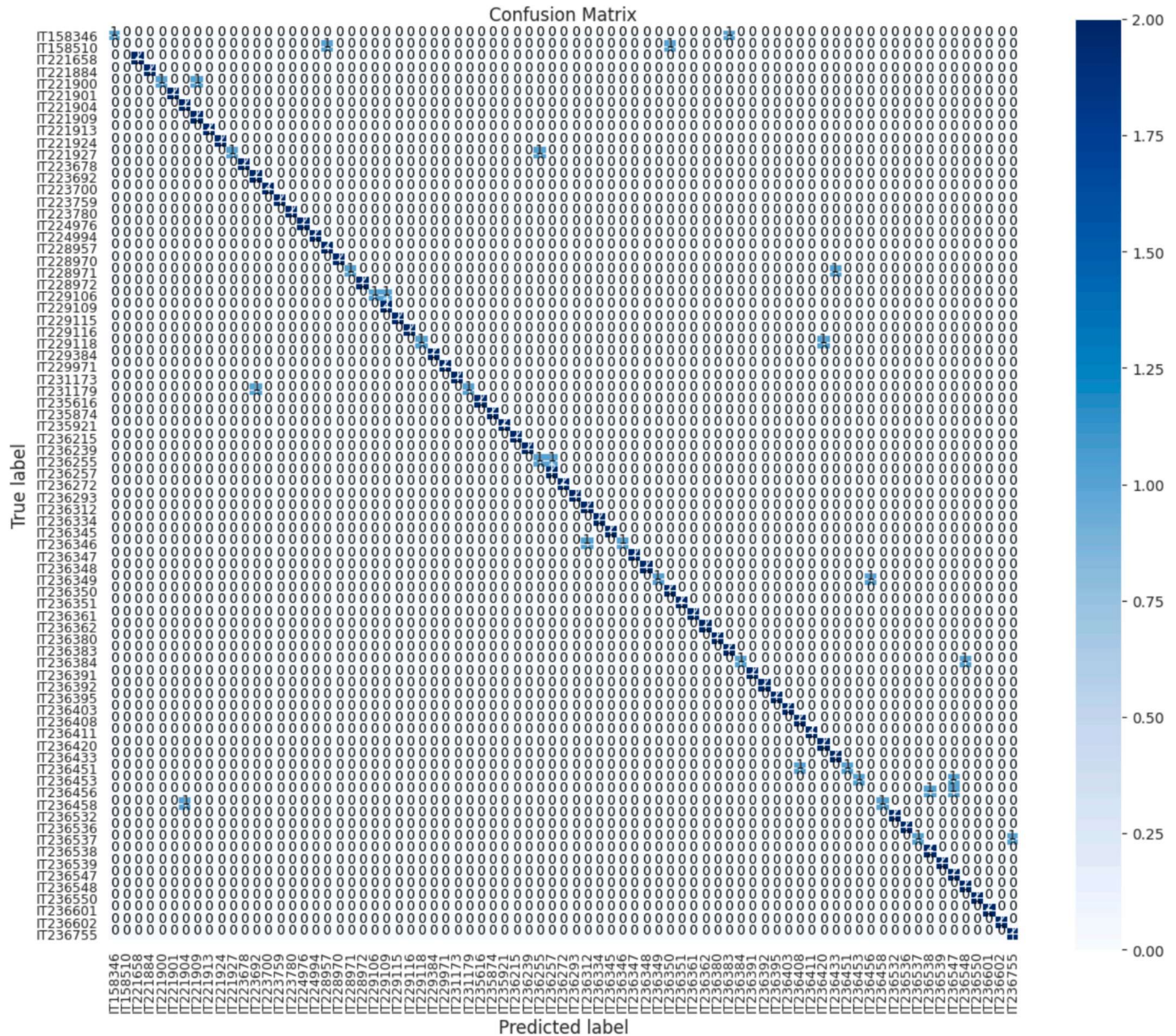


Fig. 18. Confusion Matrix of the result of the LGEfficientNetB4 method.

outperformed by the ViT method (84.58 %). The LGEfficientNetB4 method achieved an accuracy of 84.14 %, surpassing CLAEfficientNetB4 (83.70 %). These results indicate that the LeafGabor preprocessing technique is more effective for deep learning-based classification than the CLAE method. Moreover, the proposed methods significantly outperformed the baseline models, with EfficientNetB4CA reaching an accuracy of 85.46 % and LGENetB4CA achieving the highest accuracy of 85.90 %. These findings demonstrate that the EfficientNetB4CA method enhances the learning efficiency of EfficientNetB4 when applied to outdoor leaf datasets and that the LeafGabor preprocessing technique effectively highlights key features, thereby improving classification performance.

4.4. Discussions

As discussed in the introduction, deep learning techniques have shown considerable promise in plant classification, a trend that is further corroborated by the results of experiments 1 and 2, which focus on both. The findings from experiment 1 reveal that deep learning

models can achieve high accuracy in classifying chili varieties based on leaf characteristics, with accuracies ranging from 80.52 % to 96.10 % on dataset 1 and from 77.92 % to 89.61 % on dataset 2. Notably, the proposed method achieves the highest accuracy on both datasets, underscoring the potential of deep learning in capturing and classifying the geometric diversity of chili varieties. This is crucial for understanding genetic variation, advancing breeding programs, and improving crop management strategies. The proposed EfficientNetB4CA model demonstrates superior performance over the baseline EfficientNetB4, highlighting the effectiveness of incorporating the Coordinate Attention block. This architectural enhancement enables the model to better capture both spatial and channel-wise information, leading to more precise feature representations. This is especially beneficial for fine-grained classification tasks, such as chili variety identification, where subtle differences in leaf structure and texture are critical for accurate classification.

Furthermore, the LGENetB4CA model outperforms EfficientNetB4CA in both dataset 1 and dataset 2 from the JNUCLS database in experiment 1, indicating that combining the LeafGabor filter with the CA block

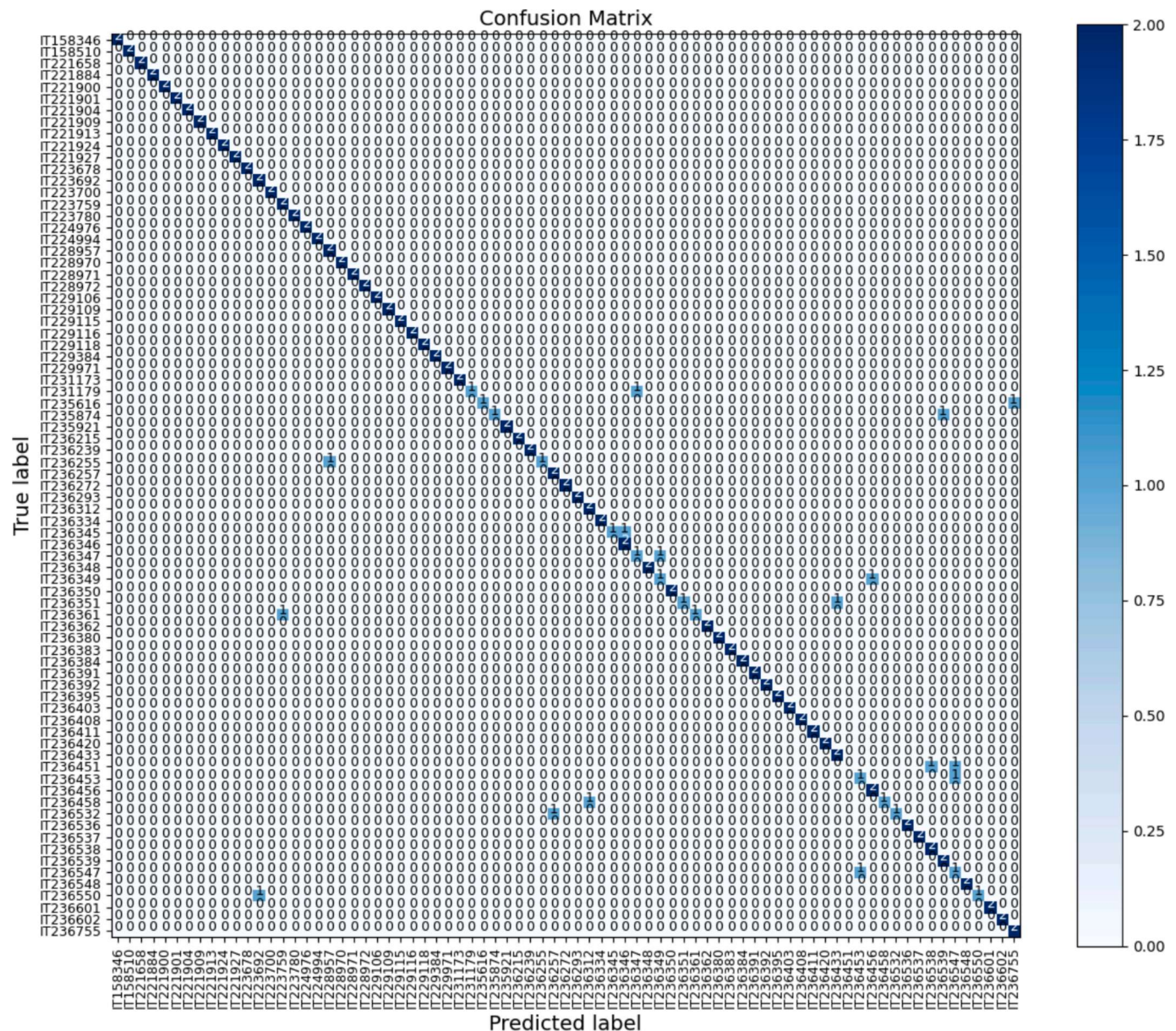


Fig. 19. Confusion Matrix of the result of the LGENetB4CA method on dataset 2.

Table 6
The comparison results of deep learning models on COLD chili dataset (%).

Methods	Precision	Recall	F1-Score	Accuracy
ViT	81.77	85.40	82.86	84.58
DenseNet121	78.32	81.02	79.02	82.82
InceptionV3	74.30	75.53	76.27	75.53
EfficientNetB3	78.28	82.18	79.67	82.38
EfficientNetB4	79.63	84.27	81.32	83.26
CLAHEEfficientNetB4 (Krishna Pratap and Suresh Kumar, 2023)	79.30	83.39	80.95	83.70
LGEfficientNetB4	80.51	84.69	82.07	84.14
EfficientNetB4CA	81.75	85.36	83.13	85.46
LGENetB4CA	82.57	87.29	84.42	85.90

further enhances the model’s ability to distinguish between closely related chili varieties. However, the benefits of combining the LeafGabor filter with EfficientNetB4CA were less pronounced in the COLD chili dataset. This may be attributed to the fact that the LeafGabor filter excels

in tasks requiring high granularity, such as variety classification; however, it, has a less significant impact on tasks focused on disease classification, where leaf shapes and structural patterns differ more substantially across classes. Despite this limitation, the findings highlight the versatility of the LeafGabor filter in improving input quality for tasks requiring fine-grained distinction, positioning it as an ideal tool for plant variety classification applications. In terms of practicality, EfficientNetB4CA also demonstrates substantial applicability. As listed in Table 7, the model has a manageable size of 215.97 MB, smaller than ViT and InceptionV3 but larger than EfficientNetB3 and DenseNet121. While its training and testing times are longer compared to other methods, its per-sample testing time remains fast (0.0571 s). This ensures that EfficientNetB4CA not only delivers high accuracy but is also efficient enough for real-world deployment in agricultural systems. Overall, the proposed method represents a significant advancement in chili variety classification, especially regarding accuracy and robustness. By integrating the CA block and LeafGabor filter, the model achieves a balance between high performance and practical feasibility. When compared to the existing literature, as outlined in Table 8, the

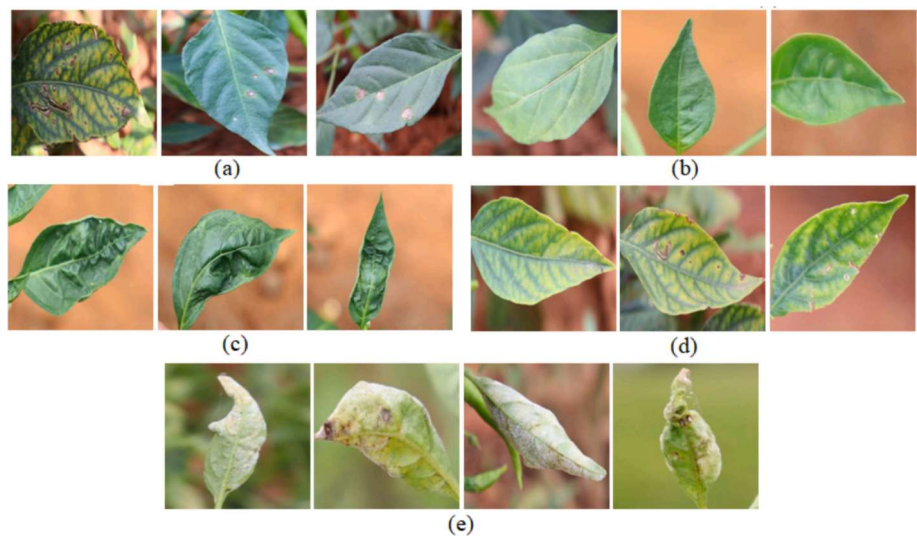


Fig. 20. Sample images from the COLD chili dataset: (a) Cercospora, (b) Healthy, (c) Mites and Thrips, (d) Nutritional Deficiency, (e) Powdery Mildew.

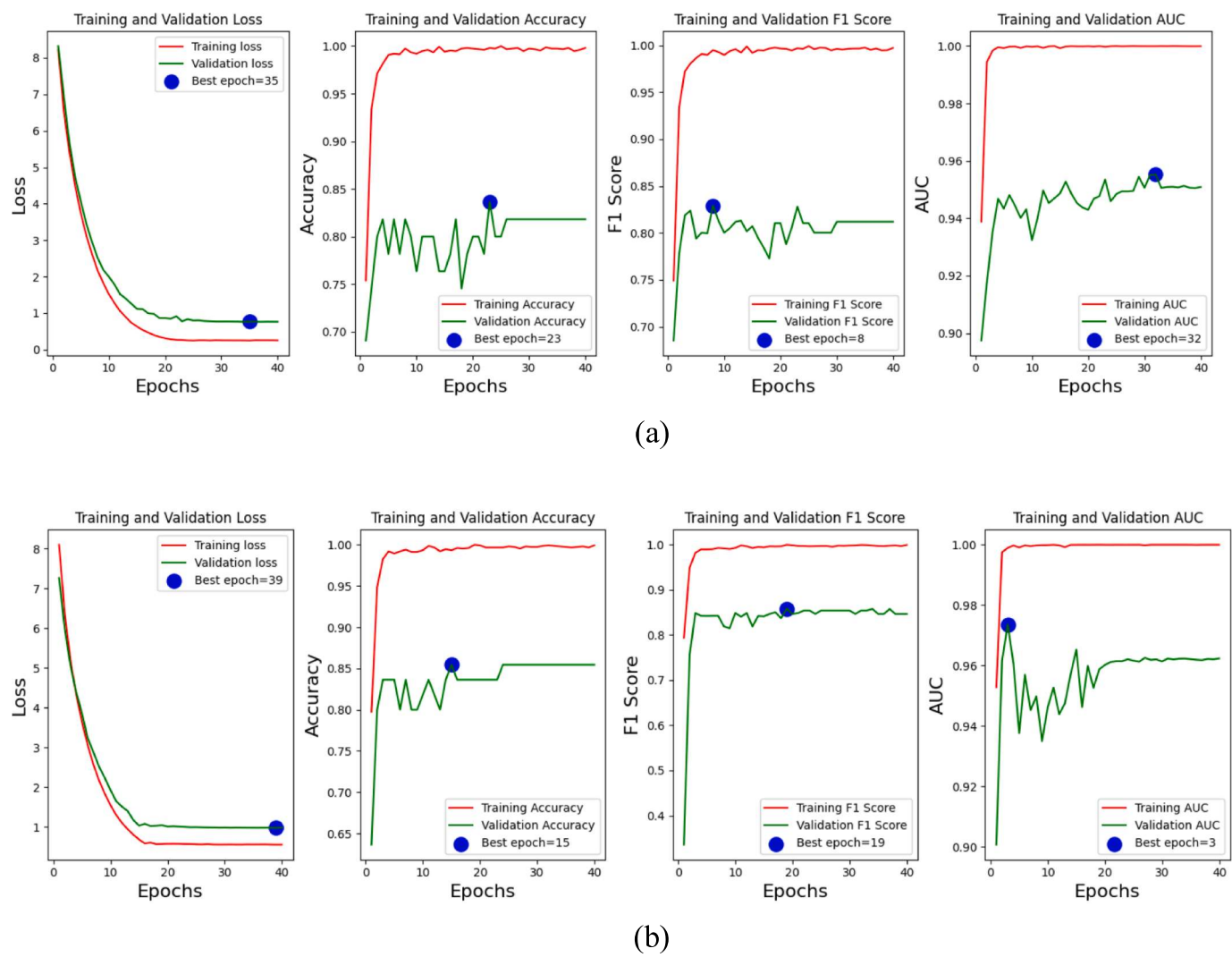


Fig. 21. A Comparison Chart of Various Methods in Training and Validation on Dataset 1: (a) EfficientNetB4 and (b) LGenetB4CA, respectively.

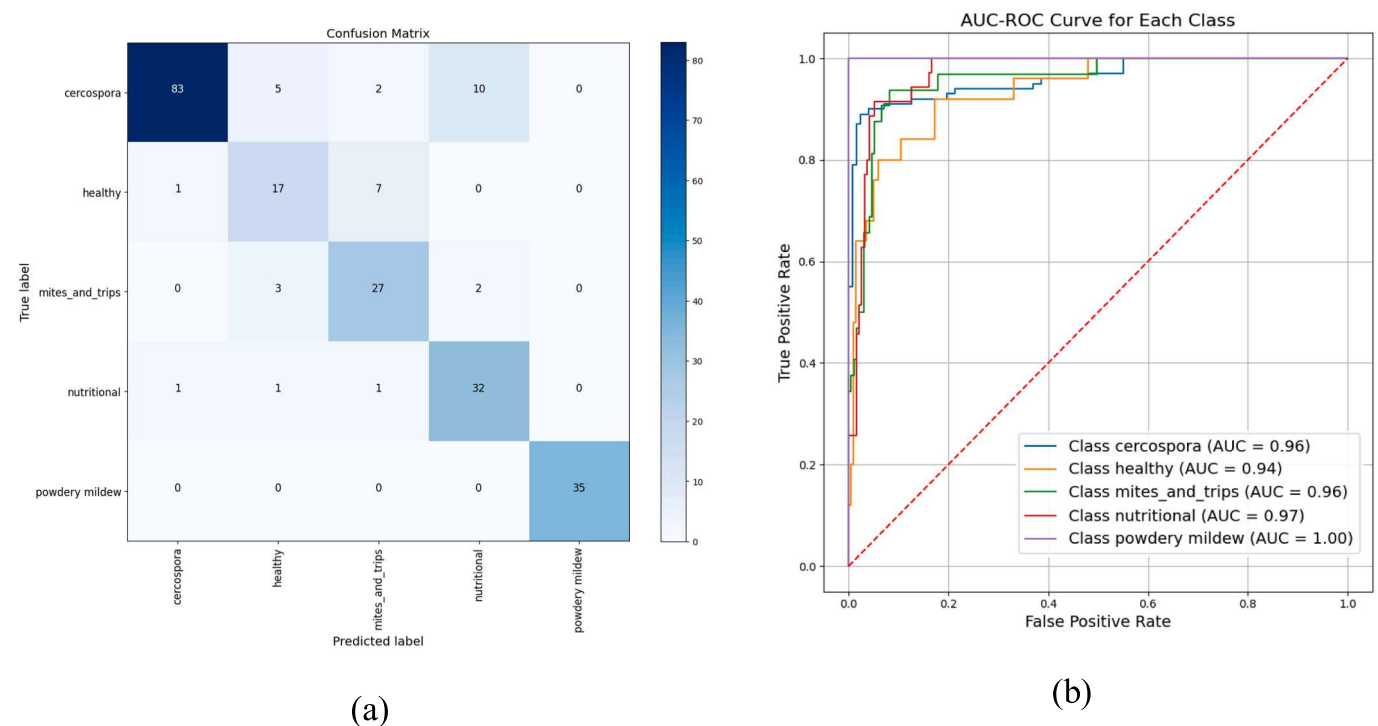


Fig. 22. (a) Confusion Matrix and (b) AUC-ROC of the EfficientNetB4CA method on COLD chili dataset.

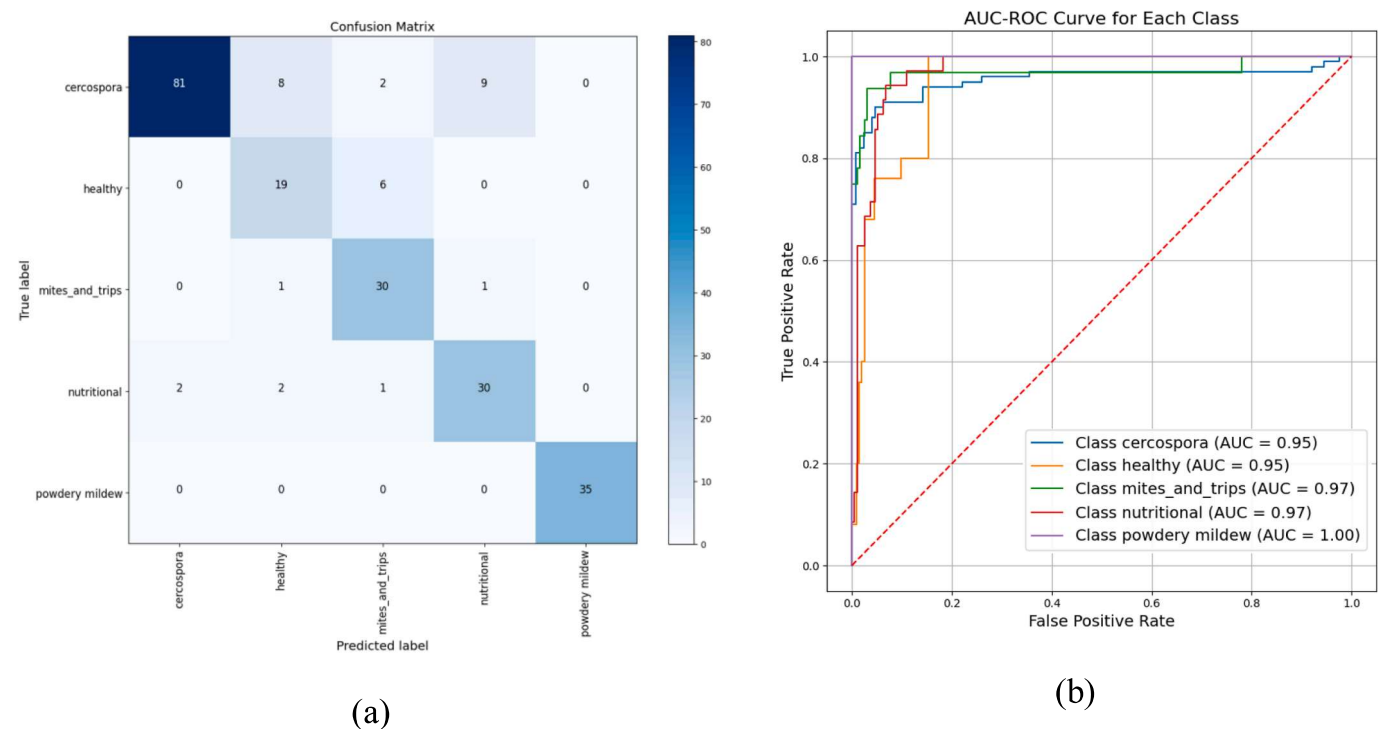


Fig. 23. (a) Confusion Matrix and (b) AUC-ROC of the LGENetB4CA method on COLD chili dataset.

results presented here mark a considerable step forward in chili classification.

The ability to recognize cultivars within a crop using AI provides several advantages for both farmers and the broader agricultural industry. For farmers, automated cultivar identification supports precise crop management by optimizing fertilizer and pesticide application, improving harvesting techniques, and minimizing post-harvest losses

through tailored storage and processing strategies. For the agricultural sector, this method enhances traceability, ensuring transparency across the supply chain, and aids in market optimization by providing insights into the regional performance of chili cultivars. Furthermore, it supports biodiversity preservation by identifying and conserving diverse cultivars, which is vital for long-term agricultural sustainability. By focusing on chili cultivar recognition, this study addresses key challenges in plant

Table 7
Comparison of size of deep learning models and prediction time.

Method	Model size (MB)	Prediction time per 1 image (s)	Training time with epoch = 20
ViT	327.54	0.0187	4h50m48s
DenseNet121	53.13	0.0126	1h19m38s
InceptionV3	256.97	0.0176	1h47m 07 s
EfficientNetB3	129.26	0.0206	3h45m20s
EfficientNetB4	208.97	0.0452	5h01m30s
EfficientNetB4CA	215.95	0.0571	5h22m08s

Table 8
Related studies summary – Plant species classification accuracy.

	Datasets	Accuracy (%)	Method
Cruz-Domínguez et al., 2021 (Cruz-Domínguez et al., 2021)	850 images, 3 classes, (Three quality levels of the Guajillo chili)	82.13 %	Grayscale based Histogram + ANN
Hendrawanet al., 2021 (Hendrawan, 2021)	1200 images, 3 classes (Three maturity levels)	93.89 %	GoogLeNet
NageswararaoNaik et al., 2022 (NageswararaoNaik et al., 2022)	5100 images, 6 classes (down curl of a leaf, Geminivirus, Cercospora leaf spot, yellow leaf disease, up curl disease, and healthy leaf)	99.12 %	SECNN
Bezabihet al., 2023 (Bezabih et al., 2023)	1596 images, 4 classes (common rust, healthy fruit, fruit rot disease, grey leaves, and healthy leaves)	95.82 %	Concatenated CNN model (VGG16 and AlexNet)
Pratapet al., 2023 (Krishna Pratap and Suresh Kumar, 2023)	2000 images, 5 classes (Healthy leaf, Up curl, Down curl, Gemini virus, Cercospora leaf spot)	95.82 %	EfficientLeafNetB4
Suwarningsihet al., 2022 (Suwarningsih et al., 2022)	1114 images, 12 classes of chili varieties	78.37 %	DenseNet-121
M. P Aishwarya et al., 2024 (Aishwarya, 2024)	COLD chili dataset: 10,987 images, 5 classes of chili disease – JNUCLS dataset: 462 images, 77 classes of chili varieties – COLD chili dataset y (Aishwarya, 2024)	79.56 % 89.61 % 85.90 %	Xception LGENetB4CA
Our proposed method			

classification, advancing precision agriculture, and promoting biodiversity conservation through innovative preprocessing and classification techniques.

Appendix A. Leaf image list of 77 chili capsicum annum germplasm

5. Conclusions

This study introduces a comprehensive leaf database that encompasses various chili cultivars to study varietal diversity based on leaf characteristics, offering innovative strategies and applications of digital image processing. Leaf samples were collected from different locations, each representing a distinct cultivar, as identified by expert biologists. We propose the LGENetB4CA method for identifying chili varieties based on leaf features. This approach involves two key steps: (1) preprocessing using the LeafGabor filter, which enhances image quality by highlighting leaf details, and (2) feature learning and classification using the EfficientNetB4CA model, which incorporates a Coordinate Attention block to capture both spatial and channel-wise information effectively.

We evaluated the proposed method against state-of-the-art deep learning models, including ViT, DenseNet121, InceptionV3, and EfficientNetB3 and B4, to benchmark its performance. Experimental results demonstrate that LGENetB4CA achieves the highest accuracy, confirming its ability to capture the geometric diversity of chili varieties based on leaf characteristics. Furthermore, the proposed method also proves effective for classifying diseases affecting chili leaves. Future research directions include expanding the database by incorporating additional chili varieties to assess the scalability and effectiveness of the proposed method with a more diverse dataset; exploring advanced preprocessing techniques or alternative deep learning architectures to enhance image quality and recognition performance; and applying the LeafGabor filter or the LGENetB4CA model to address identification challenges in other plant varieties, thereby demonstrating its versatility and broader applicability.

CRedit authorship contribution statement

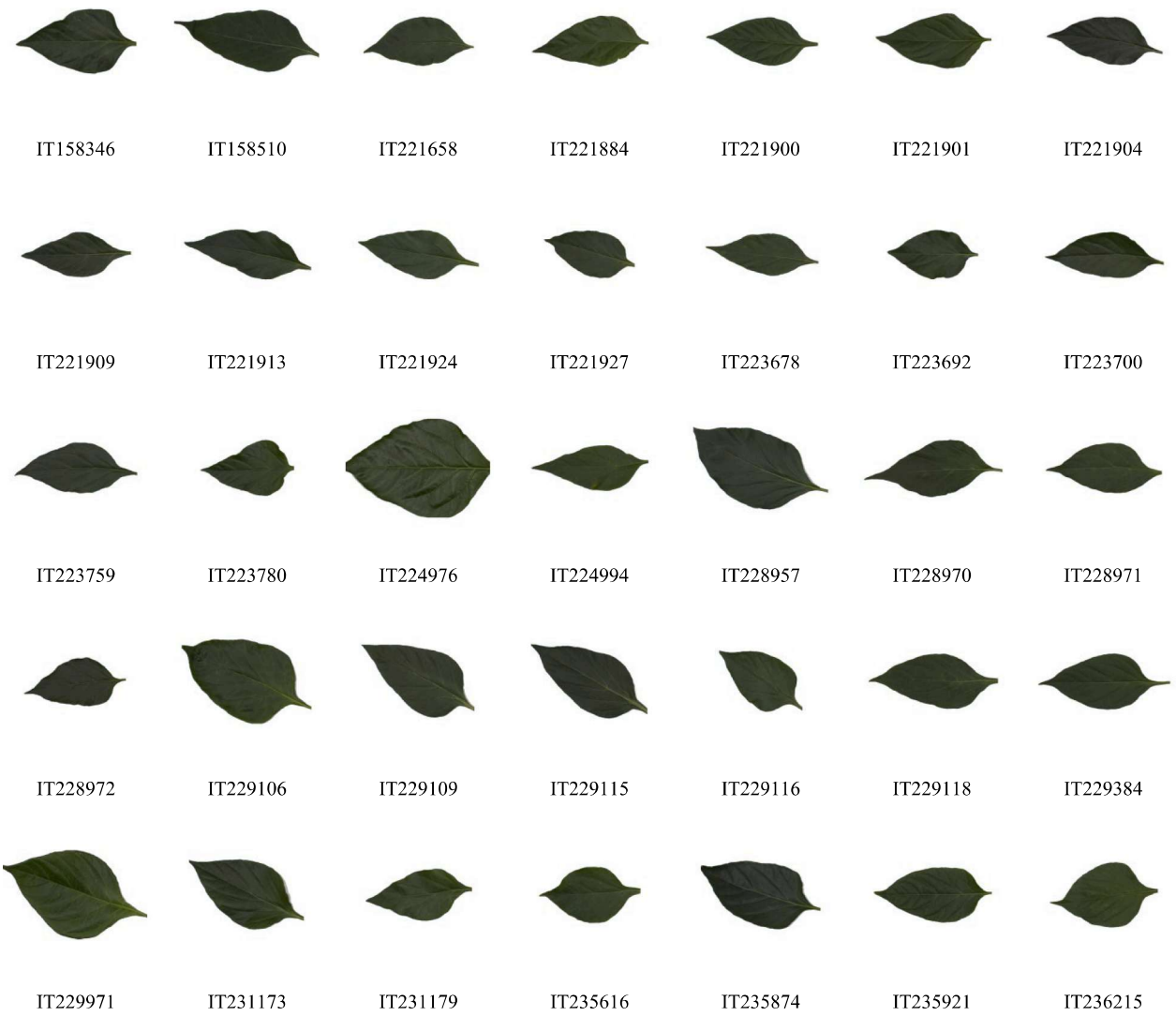
Hoang ThienVan: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis. **Giang Van Vu:** Writing – original draft, Visualization, Validation, Software, Methodology. **Thai Thanh Tuan:** Resources, Validation, Investigation. **Bay Vo:** Writing – review & editing, Validation, Supervision, Investigation. **Yong SukChung:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

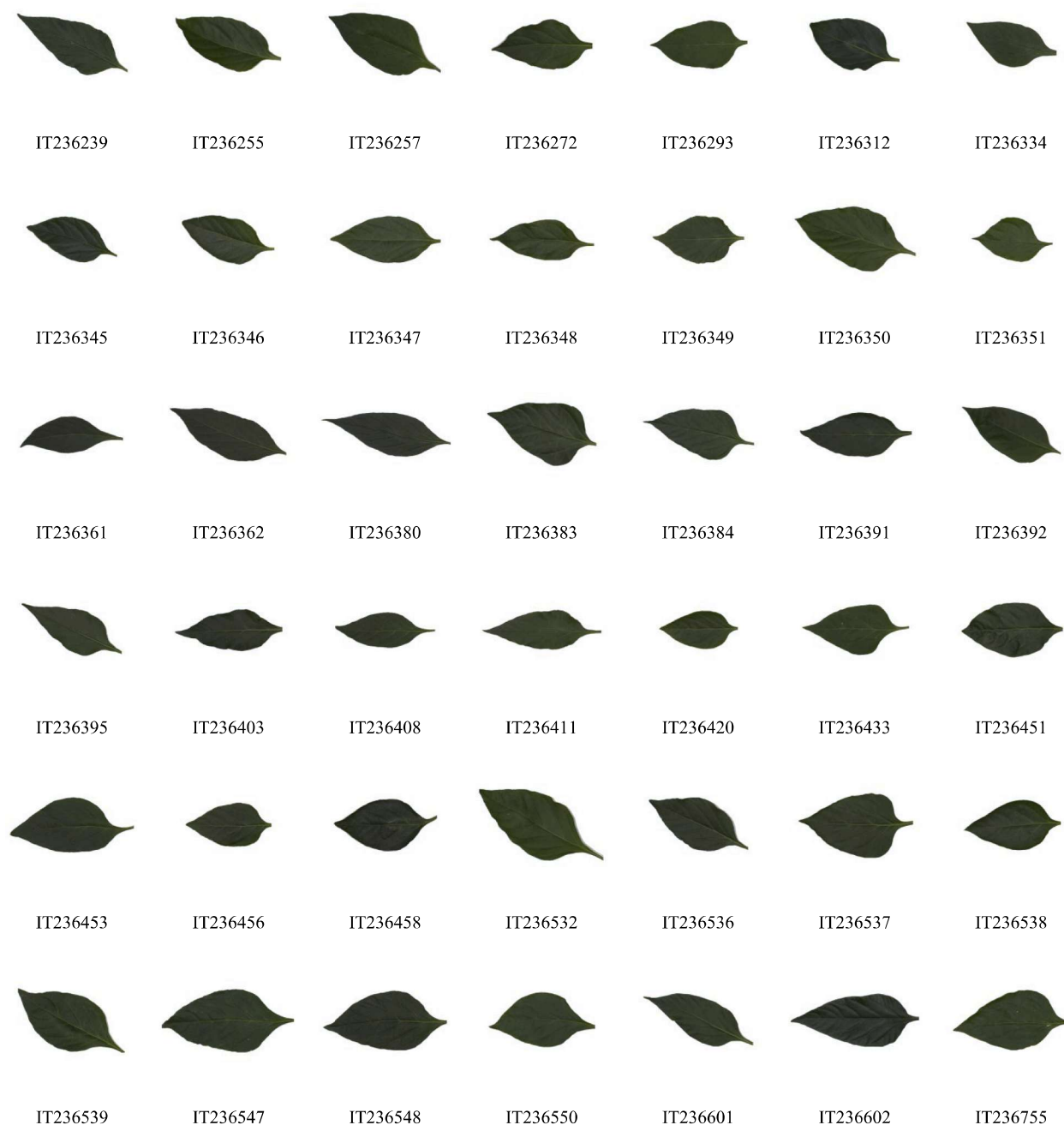
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Appendix B. Information on germplasm of chili capsicum annum

No.	varieties-id	Germplasm	Origin
1	IT158346	Perennial HDA	FRA
2	IT158510	P155/81	HND
3	IT221658	02G-130	UNK
4	IT221884	KC 857	VNM
5	IT221900	KC01309	LAO
6	IT221901	KC01310	LAO
7	IT221904	KC01315	LAO
8	IT221909	KC01323	LAO
9	IT221913	KC01327	LAO

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No.	varieties-id	Germplasm	Origin
10	IT221924	59C-161	MEX
11	IT221927	HES74P-23	USA
12	IT223678	KC 00,005	USA
13	IT223692	CMV 1166	HUN
14	IT223700	VP 2	VNM
15	IT223759	VP 87	VNM
16	IT223780	VP 117	VNM
17	IT224976	MAC MIN(OT)	VNM
18	IT224994	ACHAR	IND
19	IT228957	AC 09-062	USA
20	IT228970	BISBAS	YEM
21	IT228971	Early Jalapeno	USA
22	IT228972	Chile Pepper Institute	USA
23	IT229106	CMV 979/1	HUN
24	IT229109	CMV 984/1	HUN
25	IT229115	CMV 1024/1	HUN
26	IT229116	CMV 1026	HUN
27	IT229118	CMV 1031/1	HUN
28	IT229384	Hot chilli	VNM
29	IT229971	Peru 1-5-1	PER
30	IT231173	Red mushroom	USA
31	IT231179	Tam Jalapeno	USA
32	IT235616	WIR2597	LBV
33	IT235874	Kambi	BGR
34	IT235921	A7E0292	BGR
35	IT236215	140	BOL
36	IT236239	91	UNK
37	IT236255	9	UZB
38	IT236257	THA-PYJ-1998-77	THA
39	IT236272	53	UZB
40	IT236293	7	UNK
41	IT236312	UZB-GJG-1998-2	UZB
42	IT236334	JinyangJaerae	KOR
43	IT236345	Hot Portugal	USA
44	IT236346	San ta Fe Grand	UNK
45	IT236347	PBC413 TAM Mildjalapeno-1	USA
46	IT236348	PBC416 YJ81032	USA
47	IT236349	PBC120 HDA336	FRA
48	IT236350	PBC427 NuMex Eclipse	USA
49	IT236351	PBC428 NuMex Sunset	USA
50	IT236361	Sangsang 193F3	UNK
51	IT236362	Malang Pujon Local17	UNK
52	IT236380	Jungang Jongmyo-2000-6764	UNK
53	IT236383	Perennial HDV	UNK
54	IT236384	Wonder Hot	UNK
55	IT236391	98HES104 HuaRua	UNK
56	IT236392	98HES106 PBC30-4	UNK
57	IT236395	Jungang Jongmyo-2000-6854	UNK
58	IT236403	PBC157 HuaySithon	THA
59	IT236408	PBC586 PBC586	THA
60	IT236411	NP-51	IND
61	IT236420	B.Wonder	UNK
62	IT236433	Anjilbaeng-i	KOR
63	IT236451	Osh-kosh	RUS
64	IT236453	Bogatyri	RUS
65	IT236456	Belozerk	RUS
66	IT236458	BINNI-PUX	RUS
67	IT236532	Zumrad	UZB
68	IT236536	CCA3851	UNK
69	IT236537	ISPN2#3	UNK
70	IT236538	ISPN3#3	UNK
71	IT236539	ISPN3#7	UNK
72	IT236547	Kubanskiirannii 70/60	RUS
73	IT236548	Chinese Liant	IND
74	IT236550	WIR6235	TUR
75	IT236601	THA-SSJ-2002-16	THA
76	IT236602	CA300	THA
77	IT236755	Rajcatova paprika	CZK

Data availability

Data will be made available on request.

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